

# BALANCING REPRESENTATIONS BY IDENTIFYING AND GENERATING UNDERREPRESENTED DATA

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## ABSTRACT

Self-supervised learning inherits gaps in its source data. We study whether these gaps can be repaired while representation learning is in progress. We introduce BRIDGE, a label-free feedback loop in which the current SSL representation scores local support, a budgeted acquisition policy selects sparse but recoverable regions, and a frozen generator adds local variants before continued pretraining. The resulting intervention changes the representation used for later acquisition. Across five long-tailed and web-source regimes, BRIDGE improves average linear-probe and  $k$ NN transfer over source-matched SSL baselines. Paired full-schedule experiments further show gains with both ResNet-50 and ViT-S, and on four SSL objectives spanning contrastive, self-distillation and masked-reconstruction pretraining, while low-label fine-tuning and semantic segmentation extend the evidence beyond frozen-feature evaluation. Matched controls compare adaptive repair with one-shot and frozen policies, conventional augmentation, VLM-guided text-to-image generation, and an AIDE-style acquisition pipeline. We formalize the acquisition problem through recoverability-weighted marginal support utility. The analysis gives conditions for an interior repair region and explains why allocation priorities can change after an intervention. Isolating the loop from its components gives the clearest result: after a single repair stage, spending the budget on the densest quartile of the representation and on its extreme tail are indistinguishable, with a spread of 1.32 points across the entire score range, while the same acquisition rules separate by 8.0 points once the signal is recomputed over five stages. The benefit of targeted acquisition is therefore produced by the feedback loop and not by any individual intervention it makes.

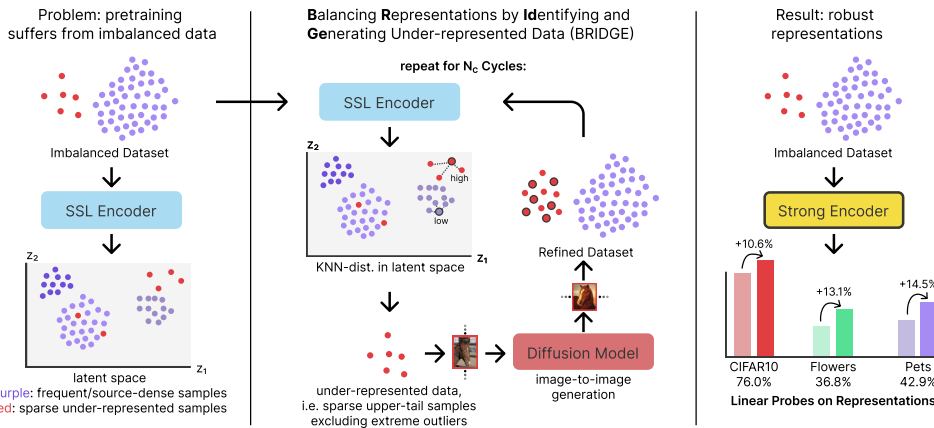


Figure 1: Conceptual overview of BRIDGE. We alternate between SSL pretraining, sparsity scoring in representation space, and targeted image-to-image diffusion augmentation. BRIDGE augments a mode-centered band inside the upper OOD tail and not the extreme tail itself.

# 1 INTRODUCTION

Self-supervised learning is a standard way to learn transferable visual representations from unlabeled images, with strong methods spanning contrastive learning, masked prediction and self-distillation (Chen et al., 2020; He et al., 2020; 2022; Caron et al., 2021; Oquab et al., 2024; Siméoni et al., 2025). They still inherit coverage biases from the source corpus: imbalance hurts self-supervised learning and has motivated specialized long-tailed training strategies (Tian et al., 2021; Assran et al., 2023; Kukleva et al., 2023; Jiang et al., 2021), and learned features concentrate around frequent modes while covering underrepresented regions weakly.

Most prior attempts at long-tailed SSL are model-centric. Temperature Schedules reshapes the optimization trajectory of contrastive learning over time (Kukleva et al., 2023) and SDCLR introduces a self-competitor to emphasize harder and rarer samples (Jiang et al., 2021). Both are strong baselines, and both treat the source distribution as fixed.

We ask instead whether the source data can be repaired while SSL training proceeds. BRIDGE, shown in Figure 1, embeds the current training set after each pretraining stage, scores each image by mean  $k$ NN distance, selects a sparse but semantically recoverable region, and generates new training images for that region with a frozen diffusion model. The next cycle trains on the repaired dataset. The encoder reveals where the source manifold is thin and the generator densifies those regions before the next round of learning.

The selector restricts attention to the upper sparse band of the OOD distribution, finds the mode within that band and selects a diverse set of anchors around it, avoiding the extreme tail of noisy images while retaining the underrepresented region most amenable to repair. BRIDGE is evaluated across five source regimes, the long-tailed ImageNet-100-LT, CIFAR-10-LT and CIFAR-100-LT and the web corpora PASS and DiffusionDB, and is best on the seven-task average in all five under both linear-probe and  $k$ NN transfer.

We make four contributions. First, we formulate data repair as recoverability-weighted support allocation and derive conditions under which a finite budget should favor sparse interior regions and update its allocation after repair, with BRIDGE as its label-free cyclic implementation. Second, we introduce an upper-tail mode-window rule with diversity-aware subsampling, targeting recoverable sparse regions without drifting into the extreme tail or spending budget on near-duplicates. Third, we evaluate five source regimes and seven transfer tasks, with paired ViT-S experiments across four SSL objectives, full fine-tuning, semantic segmentation, and matched controls for alternative feedback and generation pipelines. Fourth, we separate the acquisition rule from the loop that applies it by running the same rules under one repair stage and under five. The rules are indistinguishable under one and separate by 8.0 points under five, which locates the contribution in the repeated re-estimation and not in the selection heuristic on its own.

## 2 BACKGROUND AND RELATED WORK

**Skewed SSL and representation-driven data selection.** Long-tailed SSL methods modify training while leaving the source set fixed. SDCLR emphasizes persistently difficult samples, Temperature Schedules changes contrastive optimization, and COLT introduces external OOD data (Jiang et al., 2021; Kukleva et al., 2023; Bai et al., 2023). Other studies examine hidden imbalance, frequency-aware learning, and uncurated data (Tian et al., 2021; Assran et al., 2023; Lin et al., 2023). Data curation uses representations to retain informative subsets or remove redundancy (Whang et al., 2023; Sorscher et al., 2022; Abbas et al., 2024). Earlier augmentation methods instead add support. SMOTE interpolates labeled minority examples, while DOPING oversamples low-density latent regions for unsupervised anomaly detection (Chawla et al., 2002; Lim et al., 2018). BRIDGE uses the current SSL representation to decide where new source images should be added, then recomputes that decision after learning from them.

**Budgeted acquisition and coverage.** Active learning is the closest formal precedent for our selection rule. The core-set formulation treats acquisition as  $k$ -center coverage and selects with the same greedy farthest-first traversal we use, under the guarantee of Gonzalez (1985), though it acquires labels for existing images and creates none (Sener and Savarese, 2018); ProbCover instead maximizes covered probability mass at a fixed radius (Yehuda et al., 2022). Most relevant here, Hacohen et al.

(2022) show that the best acquisition strategy reverses with budget, dense regions being preferable while the budget is small relative to the corpus. Our single-stage sweep in Section 6.4 is consistent with that reversal. Two differences separate our setting: acquisition adds generated images and consumes no labels, so the budget is limited by generation cost, and the scoring model is the SSL encoder being trained, so each acquisition changes the function that selects the next one.

**Generative augmentation.** Diffusion models make image-conditioned repair possible through controlled denoising (Ho et al., 2020; Rombach et al., 2022; Meng et al., 2022). DA-Fusion applies this to few-shot supervised recognition and TADA targets examples learned slowly under a supervised objective (Trabucco et al., 2024; Nguyen et al., 2026), and synthetic-data utility is known to depend on prompt design, filtering, diversity and domain gap (Azizi et al., 2023; He et al., 2023; Fan et al., 2024). BRIDGE uses neither class identity nor supervised learning speed.

**Synthetic representation learning and data engines.** Synthetic imagery can serve as the full training source or define contrastive views (Ren and Lee, 2018; Baradad et al., 2021; 2022; Jahanian et al., 2022; Sariyildiz et al., 2023; Tian et al., 2023). DiffAug, the closest iterative SSL method, jointly learns a conditional generator for positive views, while BRIDGE holds the generator fixed and studies where it should intervene (Zang et al., 2024). AIDE and industrial data engines close a model–data loop from supervised task failures, using language models, annotation and domain-specific verification (Liang et al., 2024; Worker and Pathak, 2022). We evaluate language-mediated acquisition and text-to-image generation as direct alternatives.

**External representation priors.** DreamTeacher distills generative features into discriminative backbones, DIFT extracts semantic diffusion features, and REPA transfers features in the opposite direction into a diffusion model (Li et al., 2023; Tang et al., 2023; Yu et al., 2025). A frozen generator therefore contributes external knowledge even when its parameters never change, so our controls compare direct feature transfer, uniform generation and targeted generation from the same model.

### 3 THEORETICAL FRAMEWORK

We formulate targeted data repair as a budgeted allocation problem over coherent regions of representation space. A region is valuable when additional support can reduce its contribution to downstream risk, and a repair is useful only when the generator preserves the semantic content of its anchor, so the acquisition policy balances support deficit, repairability and redundancy. Let  $r_k(z; Z)$  be the distance from  $z$  to its  $k$ th neighbor in  $Z$ , and  $A$  a set of valid generated variants associated with  $z$ .

For any  $A$  we have  $r_k(z; Z \cup A) \leq r_k(z; Z)$ , strictly whenever an added point enters the first  $k$  neighbors and displaces one at strictly larger distance (Proposition ??), since enlarging a candidate set cannot increase its  $k$ th order statistic. To connect support to learning, partition representation space into coherent regions  $j$  with current effective support  $n_j$ . We model a regional learning curve by

$$\mathcal{B}(\mathbf{m}) = \sum_j p_j a_j (n_j + m_j + \kappa)^{-\gamma}, \quad (1)$$

where  $p_j$  is the transfer-relevant mass of region  $j$ ,  $a_j$  is its difficulty,  $m_j$  is the number of valid repairs allocated to it,  $\kappa > 0$ , and  $\gamma > 0$ . The marginal reduction from one valid repair,  $\Delta_j(m_j) = p_j a_j [(n_j + m_j + \kappa)^{-\gamma} - (n_j + m_j + 1 + \kappa)^{-\gamma}]$ , is positive and strictly decreasing in  $n_j + m_j$  by convexity of  $x \mapsto x^{-\gamma}$ , so between two regions of equal transfer mass and difficulty the one with less effective support has the larger marginal value (Proposition 4). A limited budget should therefore be spent away from regions that are already well supported.

Generation can fail on corrupt or incoherent anchors. Write  $d$  for a sparsity score,  $b(d)$  for marginal value conditional on valid repair,  $\rho(d)$  for the probability of a faithful variant and  $\lambda \geq 0$  for the cost of an invalid one. Expected repair utility is

$$u(d) = \rho(d)b(d) - (1 - \rho(d))\lambda. \quad (2)$$

**Proposition 1** (Interior recoverable-support optimum). *Suppose  $b$  and  $\rho$  are positive and differentiable on  $[d_{\min}, d_{\max}]$ , with  $b'(d) > 0$  and  $\rho'(d) < 0$ . Define*

$$g(d) = \frac{b'(d)}{b(d) + \lambda} + \frac{\rho'(d)}{\rho(d)}. \quad (3)$$

If  $g$  is strictly decreasing,  $g(d_{\min}) > 0$ , and  $g(d_{\max}) < 0$ , then  $u$  has a unique maximizer  $d^*$  in the interior, characterized by  $g(d^*) = 0$ .

The condition compares the relative growth of support value with the relative decay of repairability. Its monotonicity assumption on  $g$  is close to assuming the conclusion, and Proposition 5 in Appendix C discharges it for concave support value with exponentially decaying repairability, where  $d^* = \beta/\gamma$  at  $\lambda = 0$ .

The optimum moves outward when support value is closer to linear and inward when repair fidelity decays faster, which is the dependence the percentile sweeps probe. BRIDGE operationalizes the interior preference with a trimmed score band.

Consider finally a separable objective  $U(\mathbf{m}) = \sum_j U_j(n_j + m_j)$  with each  $U_j$  increasing and discretely concave and  $\sum_j m_j = B$ .

**Proposition 2** (Adaptive allocation under diminishing returns). *For integer allocations, repeatedly assigning one repair to the region with the largest current marginal gain maximizes  $U(\mathbf{m})$ . A fixed allocation can become suboptimal after a region’s marginal gain falls below that of another region.*

Recomputing the acquisition signal lets the policy follow these changing values, whereas a frozen policy keeps spending according to its initial ordering. The same model bounds what any acquisition rule can achieve when applied once.

**Proposition 3** (Single-stage insensitivity). *Let  $\mathbf{m}$  and  $\mathbf{m}'$  be any two allocations of the same budget  $B$  under Equation 1, computed from the same representation and applied without re-estimation. Then*

$$|\mathcal{B}(\mathbf{m}) - \mathcal{B}(\mathbf{m}')| \leq B \gamma \max_j p_j a_j (n_j + \kappa)^{-\gamma-1}. \quad (4)$$

*The bound tends to zero as  $B/n_{\min} \rightarrow 0$ , where  $n_{\min}$  is the least support any region carries.*

Under one intervention the choice of acquisition rule is therefore bounded by the ratio of budget to existing support, and no rule separates from any other when that ratio is small. Section 6.4 measures 2,500 generated images against a source of roughly 10,000 and finds every score band within 1.32 points. The separation over five stages is outside the bound, because re-estimation makes the support entering the score depend on earlier allocations, and we claim no matching lower bound for the repeated regime. Within a candidate region, farthest-first traversal gives a factor-two approximation to metric  $k$ -center coverage (Gonzalez, 1985).

A practical acquisition policy should therefore estimate local support, avoid sparse regions whose samples cannot be repaired faithfully, cover the chosen region without redundant anchors, and recompute its allocation after each intervention. Appendix C carries the proofs. Equation 1 and the monotonic behavior assumed for  $b$  and  $\rho$  are modeling assumptions, and their observable predictions are tested by the controls in Section 6.

## 4 BRIDGE

### 4.1 CYCLIC ACQUISITION AND CONTINUED PRETRAINING

BRIDGE instantiates each requirement without labels. Mean  $k$ NN distance estimates local support, an upper-tail mode window seeks a sparse region while excluding the extreme scores where faithful repair is less reliable, farthest-point sampling limits redundancy within it, and image-to-image diffusion supplies local variants. Continued SSL changes the representation, after which the budget is allocated again. Each cycle continues from the previous encoder and never retrains from scratch, so a cycle costs one pass over the grown source set and not a full pretraining run.

Let  $D^{(0)}$  be an unlabeled source dataset and  $f_\theta$  an SSL encoder. BRIDGE alternates pretraining and repair over  $C$  cycles; at cycle  $c$  the encoder is  $f_{\theta^{(c)}}$ , the source set  $D^{(c)}$  and its embedding set  $Z^{(c)}$ .

Three operators run after each training stage: score embeds the current source set and assigns an unlabeled sparsity score, select turns that distribution into an anchor set under a generation budget, and repair conditions the generator on those anchors and appends the variants. The encoder is the only learned component, so BRIDGE needs no labels and works with any SSL objective that provides embeddings.

After training on  $D^{(c)}$  we embed every image,  $z_i^{(c)} = f_{\theta^{(c)}}(x_i)$ , and compute a nonparametric OOD score from mean  $k$ NN distance,

$$d_i^{(c)} = \frac{1}{k} \sum_{z \in \mathcal{N}_k(z_i^{(c)}; Z^{(c)} \setminus \{z_i^{(c)}\})} \|z_i^{(c)} - z\|_2^2, \quad (5)$$

where  $\mathcal{N}_k(z; Z)$  denotes the  $k$  nearest neighbors of  $z$  in  $Z$ , and large  $d_i^{(c)}$  indicates an image in a sparse part of the learned representation space. Deep nearest-neighbor distance is also effective for OOD detection (Sun et al., 2022).

#### 4.2 SPARSE AND RECOVERABLE ANCHOR SELECTION

Selecting the largest OOD scores directly is unstable, because the farthest samples are often artifacts or incoherent outliers, while uniform sampling spreads budget over regions already well covered. We target a sparse region that is still internally coherent, in three steps. First, the selector restricts attention to scores between the 75th and 99th percentiles and builds a histogram over that band, whose modal bin has center  $m^{(c)}$ . Second, given an augmentation budget  $B$  and a candidate-pool multiplier  $\alpha$ , it forms the mode-centered candidate set of size  $M = \alpha B$  that minimizes  $\sum_{i \in C} |d_i^{(c)} - m^{(c)}|$  over  $C \subseteq U^{(c)}$  with  $|C| = M$ , where  $U^{(c)} = \{i : q_{0.75}^{(c)} \leq d_i^{(c)} \leq q_{0.99}^{(c)}\}$ . Third, the anchor set is  $S^{(c)} = \text{FPS}(C^{(c)}, B; Z^{(c)})$ , greedy farthest-point sampling in embedding space from that pool. This keeps selection inside a stable sparse regime while discouraging near-duplicate anchors. We use  $\alpha = 4$ . Appendix Figures 4–5 show where the budget lands and how the acquisition distribution moves across cycles.

#### 4.3 LOCAL GENERATIVE REPAIR

Given  $S^{(c)}$ , we generate  $G$  variants per selected image with a frozen image-to-image diffusion model  $g_\phi$ , conditioning on the source image with an empty text prompt so the repair signal comes from the anchor and not from class names, and set  $D^{(c+1)} = D^{(c)} \cup \{g_\phi(x_i, \epsilon_j) : i \in S^{(c)}, j \in [G]\}$ . The noise  $\epsilon_j$  indexes independent translations of one anchor, so variants keep coarse structure while changing texture, pose, color or composition. No generated-image filter is applied. The main method uses Stable Diffusion 3 Medium in image-to-image mode with 20 denoising steps, guidance scale 5.0, strength 0.6, and output resolution equal to the training crop size. With an empty prompt the guidance scale is inert: at a fixed seed, scales 3.0, 5.0 and 7.5 give bit-identical images and scale 1.0 differs by a mean absolute pixel error of 0.0016 at 49.7 dB. Strength sets how far a variant may travel from its anchor. The generation ablation replaces this operator with FLUX at six steps and guidance scale 2.5, or with oracle re-population, which re-adds the originally removed real source images without diffusion and is the upper bound on what any repair operator can achieve at this budget.

These components implement the allocation principle of Section 3: the score estimates support deficit, the mode window is a recoverability constraint, FPS approximates coverage, and cyclic re-estimation updates the allocation after every intervention.

### 5 EXPERIMENTAL SETUP

**Source datasets.** We pretrain on five unlabeled source regimes: the long-tailed classification datasets ImageNet-100-LT (Liu et al., 2019), CIFAR-10-LT and CIFAR-100-LT (Krizhevsky, 2009), and 10k subsets of the web corpora PASS (Asano et al., 2021) and DiffusionDB (Wang et al., 2023).

**Downstream evaluation.** Transfer is evaluated on seven benchmarks: CIFAR-10-LT, CIFAR-100-LT, Stanford Cars (Krause et al., 2013), FGVC Aircraft (Maji et al., 2013), Oxford Flowers (Nilsback and Zisserman, 2008), Oxford-IIIT Pets (Parkhi et al., 2012) and ImageNet-100-LT. The main text reports linear probes with  $k$ NN corroboration; the appendix expands both.

**Training protocol.** Reported numbers are mean  $\pm$  standard deviation over three seeds. ResNet-50 is the default backbone, the common architecture across the baselines we compare against, and an architecture ablation covers ResNet-18 through ViT-B. BRIDGE uses  $k = 100$ , selects 500 images

Table 1: Linear-probe transfer for the three label-derived long-tailed source regimes. The average is computed over all seven downstream linear-probe tasks.

| Source          | Method        | C10                              | C100                             | Cars                             | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|-----------------|---------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| ImageNet-100-LT | SimCLR        | 73.9 $\pm$ 0.5                   | 46.8 $\pm$ 0.4                   | 13.7 $\pm$ 0.3                   | 19.6 $\pm$ 0.2                   | 35.9 $\pm$ 7.2                   | 38.6 $\pm$ 1.5                   | 50.4 $\pm$ 0.5                   | 39.8 $\pm$ 1.5                   |
|                 | TS            | 72.5 $\pm$ 0.4                   | 44.0 $\pm$ 0.5                   | 12.5 $\pm$ 1.2                   | 15.0 $\pm$ 0.1                   | 35.7 $\pm$ 1.8                   | 40.6 $\pm$ 2.2                   | <b>56.7 <math>\pm</math> 1.2</b> | 39.6 $\pm$ 1.1                   |
|                 | SDCLR         | 41.1 $\pm$ 1.7                   | 18.0 $\pm$ 0.6                   | 1.4 $\pm$ 0.3                    | 2.3 $\pm$ 0.5                    | 10.8 $\pm$ 1.7                   | 10.4 $\pm$ 0.3                   | 21.9 $\pm$ 0.6                   | 15.1 $\pm$ 0.8                   |
|                 | BRIDGE (ours) | <b>75.8 <math>\pm</math> 0.2</b> | <b>49.3 <math>\pm</math> 0.2</b> | <b>16.4 <math>\pm</math> 1.2</b> | <b>22.4 <math>\pm</math> 0.4</b> | <b>36.8 <math>\pm</math> 1.9</b> | <b>42.9 <math>\pm</math> 1.0</b> | 53.8 $\pm$ 0.1                   | <b>42.5 <math>\pm</math> 0.7</b> |
| CIFAR-10-LT     | SimCLR        | 67.8 $\pm$ 0.2                   | 38.0 $\pm$ 1.0                   | 6.5 $\pm$ 0.4                    | 16.0 $\pm$ 0.9                   | <b>19.2 <math>\pm</math> 1.6</b> | 24.0 $\pm$ 1.3                   | 32.0 $\pm$ 0.7                   | 29.1 $\pm$ 0.9                   |
|                 | TS            | 68.5 $\pm$ 1.3                   | 36.0 $\pm$ 1.7                   | 4.3 $\pm$ 0.2                    | 10.3 $\pm$ 0.1                   | 13.1 $\pm$ 2.0                   | 19.7 $\pm$ 2.3                   | 31.5 $\pm$ 1.2                   | 26.2 $\pm$ 1.3                   |
|                 | SDCLR         | 39.6 $\pm$ 1.1                   | 15.6 $\pm$ 0.2                   | 1.1 $\pm$ 0.1                    | 2.4 $\pm$ 0.9                    | 12.0 $\pm$ 1.6                   | 9.1 $\pm$ 1.8                    | 16.5 $\pm$ 1.6                   | 13.8 $\pm$ 1.1                   |
|                 | BRIDGE (ours) | 78.1 $\pm$ 0.3                   | <b>49.8 <math>\pm</math> 0.4</b> | <b>9.2 <math>\pm</math> 0.8</b>  | <b>17.7 <math>\pm</math> 1.0</b> | 18.2 $\pm$ 1.1                   | <b>27.4 <math>\pm</math> 2.7</b> | 39.0 $\pm$ 0.6                   | <b>34.2 <math>\pm</math> 1.0</b> |
| CIFAR-100-LT    | SimCLR        | 59.0 $\pm$ 1.5                   | 31.1 $\pm$ 1.6                   | 4.1 $\pm$ 0.3                    | 11.2 $\pm$ 1.0                   | 12.6 $\pm$ 3.4                   | 20.1 $\pm$ 1.0                   | 28.7 $\pm$ 1.3                   | 23.8 $\pm$ 1.4                   |
|                 | TS            | 56.3 $\pm$ 0.2                   | 28.0 $\pm$ 0.1                   | 2.8 $\pm$ 0.6                    | 8.5 $\pm$ 0.2                    | 15.5 $\pm$ 2.5                   | 16.2 $\pm$ 3.1                   | 26.2 $\pm$ 1.8                   | 21.9 $\pm$ 1.2                   |
|                 | SDCLR         | 37.1 $\pm$ 0.8                   | 15.7 $\pm$ 0.2                   | 1.9 $\pm$ 0.1                    | 2.4 $\pm$ 0.7                    | 9.4 $\pm$ 1.0                    | 9.9 $\pm$ 1.7                    | 16.2 $\pm$ 0.3                   | 13.2 $\pm$ 0.7                   |
|                 | BRIDGE (ours) | <b>74.2 <math>\pm</math> 0.5</b> | <b>48.1 <math>\pm</math> 0.4</b> | <b>8.3 <math>\pm</math> 1.0</b>  | <b>18.2 <math>\pm</math> 0.7</b> | <b>24.4 <math>\pm</math> 2.3</b> | 27.6 $\pm$ 0.7                   | 39.8 $\pm$ 1.1                   | <b>34.4 <math>\pm</math> 1.0</b> |

per cycle, generates 5 per selected image and trains 5 cycles of 100 epochs; the cycle ablation varies the cycle count while adjusting the per-cycle budget so that total volume is held fixed. DINO follows the published reference recipe with a ViT-S backbone, eight local crops at  $96 \times 96$  and a bottleneck projection head, and is not run on ResNet-50 because that recipe is defined for ViT backbones. MAE is likewise ViT-only and runs five cycles of 80 epochs. Because the downstream tasks are heterogeneous we emphasize per-dataset tables and use the average as a summary statistic. Appendix D gives implementation details, runtime summaries and asset notes.

**Compared methods and ablations.** We compare SimCLR, TS, SDCLR and BRIDGE, with hybrids in the appendix. All ablations run on ImageNet-100-LT source pretraining with the same downstream suite and three seeds, varying one component at a time: the SSL objective, the generator, the selection rule, the cycle count and the backbone. The generation ablation includes oracle re-population, which returns the exact images withheld when the long tail was constructed and places real support where the method only estimates that support is missing. That arm is the ceiling available to any repair operator at this budget, so a generator approaching it is performing as well as the data allow.

## 6 RESULTS

### 6.1 COMPARISON ACROSS SOURCE DATASETS

Table 1 reports the three label-derived long-tailed source regimes with all seven downstream tasks, and BRIDGE is strongest on the overall average in all three: under CIFAR-100-LT it raises the seven-task average from 23.8 for source-matched SimCLR to 34.4 and is best on every dataset, on ImageNet-100-LT from 39.8 to 42.5 with TS competitive only in domain, and on CIFAR-10-LT from 29.1 to 34.2.

The ranking is not specific to linear probing: under  $k$ NN BRIDGE remains strongest on average in all three regimes, improving over SimCLR from 31.8 to 33.6 on ImageNet-100-LT, 24.0 to 28.7 on CIFAR-10-LT and 20.9 to 27.0 on CIFAR-100-LT.

Table 2 evaluates two web-source regimes: PASS is a natural web image corpus and DiffusionDB is fully synthetic. BRIDGE is again strongest on average in both, improving the seven-task average over source-matched SimCLR from 36.8 to 39.9 on PASS-10k and from 37.7 to 40.2 on DiffusionDB-10k. Because the DiffusionDB source data are already synthetic, the gains cannot be attributed to harvesting more natural web diversity, and the repair loop remains useful when the source corpus itself carries strong generative and stylistic bias.

Under  $k$ NN the same holds, 29.3 to 30.6 on PASS-10k and 28.2 to 29.9 on DiffusionDB-10k, placing the effect outside both the learned probe and the natural-image setting.

### 6.2 EVALUATION BEYOND THE DEFAULT PROTOCOL

Table 3 tests whether the result is specific to SimCLR with ResNet-50 or to frozen-feature evaluation. Four objectives are run paired on ViT-S under one schedule and one repair budget, covering

Table 2: Linear-probe transfer for the two web-source regimes. PASS is a natural web-image corpus, while DiffusionDB is fully synthetic. The average is computed over all seven downstream linear-probe tasks.

| Source          | Method        | C10                              | C100                             | Cars                             | Aircraft                         | Flowers                          | Pets                             | IN100-LT       | Avg.                             |
|-----------------|---------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------|----------------------------------|
| PASS-10k        | SimCLR        | 72.3 $\pm$ 0.2                   | 45.5 $\pm$ 0.5                   | 11.0 $\pm$ 0.2                   | 17.5 $\pm$ 1.4                   | 33.3 $\pm$ 1.3                   | 33.3 $\pm$ 0.9                   | 44.7 $\pm$ 0.9 | 36.8 $\pm$ 0.8                   |
|                 | TS            | 71.5 $\pm$ 0.2                   | 43.5 $\pm$ 0.4                   | 10.2 $\pm$ 1.2                   | 17.2 $\pm$ 0.2                   | 33.1 $\pm$ 2.1                   | 36.4 $\pm$ 1.8                   | 49.4 $\pm$ 0.4 | 37.3 $\pm$ 0.9                   |
|                 | SDCLR         | 42.2 $\pm$ 1.1                   | 17.7 $\pm$ 0.8                   | 1.8 $\pm$ 0.3                    | 2.2 $\pm$ 0.8                    | 10.6 $\pm$ 1.3                   | 8.2 $\pm$ 1.6                    | 20.8 $\pm$ 0.4 | 14.8 $\pm$ 0.9                   |
|                 | BRIDGE (ours) | <b>73.9 <math>\pm</math> 0.3</b> | <b>47.1 <math>\pm</math> 0.4</b> | <b>13.6 <math>\pm</math> 0.7</b> | <b>20.5 <math>\pm</math> 0.7</b> | <b>37.7 <math>\pm</math> 4.0</b> | <b>37.1 <math>\pm</math> 0.7</b> | 49.1 $\pm$ 0.5 | <b>39.9 <math>\pm</math> 1.0</b> |
| DiffusionDB-10k | SimCLR        | 74.4 $\pm$ 0.5                   | 48.6 $\pm$ 0.2                   | 10.9 $\pm$ 0.7                   | 17.6 $\pm$ 0.8                   | 31.2 $\pm$ 2.4                   | 36.1 $\pm$ 1.3                   | 45.3 $\pm$ 0.5 | 37.7 $\pm$ 0.9                   |
|                 | TS            | 72.3 $\pm$ 0.3                   | 44.4 $\pm$ 0.3                   | 10.7 $\pm$ 0.6                   | 15.4 $\pm$ 0.6                   | 33.1 $\pm$ 1.1                   | <b>39.9 <math>\pm</math> 2.7</b> | 48.8 $\pm$ 0.2 | 37.8 $\pm$ 0.8                   |
|                 | SDCLR         | 42.4 $\pm$ 0.2                   | 18.4 $\pm$ 0.2                   | 1.1 $\pm$ 0.2                    | 2.4 $\pm$ 0.1                    | 13.0 $\pm$ 3.7                   | 8.2 $\pm$ 1.4                    | 21.0 $\pm$ 0.6 | 15.2 $\pm$ 0.9                   |
|                 | BRIDGE (ours) | <b>75.9 <math>\pm</math> 0.4</b> | <b>49.9 <math>\pm</math> 0.5</b> | <b>12.8 <math>\pm</math> 1.0</b> | <b>20.0 <math>\pm</math> 0.4</b> | <b>35.7 <math>\pm</math> 1.1</b> | 37.7 $\pm$ 0.8                   | 49.1 $\pm$ 0.5 | <b>40.1 <math>\pm</math> 0.7</b> |

Table 3: Evaluation beyond the default SimCLR/ResNet-50 linear-probe setting. Panel A reports linear-probe accuracy on four of the seven downstream datasets for five paired ViT-S objectives, three seeds each; Table 24 carries all seven and Table 25 the  $k$ NN protocol. Panel B reports three paired seeds for low-label full fine-tuning and PASCAL VOC semantic segmentation.

| A. Frozen ViT-S transfer |              |                                   |                                    |                                    | B. End-to-end downstream adaptation |                  |                      |                  |                  |
|--------------------------|--------------|-----------------------------------|------------------------------------|------------------------------------|-------------------------------------|------------------|----------------------|------------------|------------------|
| Objective                | Method       | Cars                              | Aircraft                           | Flowers                            | IN100-LT                            | Evaluation       | Setting              | SSL baseline     | With BRIDGE      |
| SimCLR                   | SSL baseline | 7.15 $\pm$ 0.19                   | 10.92 $\pm$ 0.40                   | 32.90 $\pm$ 2.72                   | 41.95 $\pm$ 1.43                    | Full fine-tuning | IN100-LT, 1% labels  | 14.31 $\pm$ 0.33 | 16.29 $\pm$ 1.11 |
|                          | With BRIDGE  | <b>8.18 <math>\pm</math> 0.42</b> | <b>11.82 <math>\pm</math> 0.43</b> | <b>36.60 <math>\pm</math> 2.85</b> | <b>43.93 <math>\pm</math> 1.09</b>  |                  | IN100-LT, 10% labels | 21.56 $\pm$ 0.75 | 24.72 $\pm$ 1.75 |
| MoCo v3                  | SSL baseline | 7.57 $\pm$ 0.23                   | 11.05 $\pm$ 1.31                   | 30.28 $\pm$ 1.36                   | 44.79 $\pm$ 1.17                    | Full fine-tuning | PASCAL VOC, mIoU     | 12.78 $\pm$ 0.79 | 15.20 $\pm$ 0.86 |
|                          | With BRIDGE  | <b>9.55 <math>\pm</math> 0.34</b> | <b>13.42 <math>\pm</math> 0.98</b> | <b>36.82 <math>\pm</math> 6.07</b> | <b>47.55 <math>\pm</math> 0.86</b>  |                  |                      |                  |                  |
| DINO                     | SSL baseline | 8.10 $\pm$ 0.25                   | 11.40 $\pm$ 1.03                   | <b>38.13 <math>\pm</math> 5.24</b> | 44.63 $\pm$ 1.47                    | Segmentation     |                      |                  |                  |
|                          | With BRIDGE  | <b>9.49 <math>\pm</math> 0.88</b> | <b>13.28 <math>\pm</math> 0.58</b> | <b>37.25 <math>\pm</math> 2.61</b> | <b>46.78 <math>\pm</math> 0.77</b>  |                  |                      |                  |                  |
| MAE                      | SSL baseline | 1.54 $\pm$ 0.19                   | 2.44 $\pm$ 1.00                    | <b>10.46 <math>\pm</math> 5.88</b> | 19.93 $\pm$ 2.13                    |                  |                      |                  |                  |
|                          | With BRIDGE  | <b>1.56 <math>\pm</math> 0.43</b> | <b>3.18 <math>\pm</math> 0.70</b>  | 8.50 $\pm$ 2.26                    | <b>22.19 <math>\pm</math> 2.35</b>  |                  |                      |                  |                  |

contrastive learning (SimCLR), momentum contrast (MoCo v3), self-distillation (DINO) and masked reconstruction (MAE). Counting linear-probe and  $k$ NN accuracy on all seven downstream datasets gives 14 measurements per objective. BRIDGE improves 12 of 14 for SimCLR, 12 of 14 for MoCo v3, 13 of 14 for DINO and 12 of 14 for MAE, with mean changes of +1.32, +1.31, +1.88 and +1.38 points and a two-sided sign test at  $p \leq .013$  in each case. The largest single gains are on Pets, +7.06 points for DINO and +4.80 for SimCLR under linear probing. Under  $k$ NN, MoCo v3 loses 3.88 and 3.74 points on CIFAR-10 and CIFAR-100 while gaining on the other five datasets.

The same source representations improve after end-to-end adaptation. With 1 and 10 percent of the ImageNet-100-LT labels, full fine-tuning gains  $1.97 \pm 0.78$  and  $3.17 \pm 1.66$  points. PASCAL VOC semantic segmentation improves from  $12.78 \pm 0.79$  to  $15.20 \pm 0.86$  mIoU, a paired gain of  $2.42 \pm 0.27$  over three seeds.

### 6.3 ABLATIONS

**SSL objective and backbone.** Swapping the objective on a fixed ResNet-50 confounds it with an architecture mismatch, since self-distillation methods are normally reported with ViT backbones. The appendix retains the ResNet-50 comparison for SimCLR and MoCo, and the comparison involving DINO and MAE runs on ViT-S with the published recipes, so each arm is evaluated in the pairing its authors intended and no row in this paper reports a self-distillation objective on a convolutional backbone.

**Generation mechanism.** Under matched sampling settings the three-seed rerun no longer supports a ranking between FLUX.1-schnell and SD3, so generator choice is an implementation choice. Oracle re-population is an upper bound with access to information the method does not have, the identity and location of the missing support, so repair coming close to it recovers most of the benefit of the withheld real images without access to them, and the gap is the cost of estimating where support is missing.

**Is the gain transfer from the generator?** A frozen generator trained on external data carries information the source corpus does not, so the improvement could reflect distillation from it and not the acquisition rule. The selection ablation holds the generator fixed: its five arms call the same Stable Diffusion 3 checkpoint at the same settings, add the same number of images and receive the same updates, and only the conditioning anchors vary. An account in which the generator supplies

the benefit predicts no spread there, yet uniform reaches  $35.5 \pm 1.2$  and top-tail  $34.3 \pm 1.2$  against 42.1 to 42.3 for the three mode windows, every one ahead after Holm correction at adjusted  $p \leq .011$ . Those 6.6 to 8.0 points are attributable to the placement of the budget. The appendix reports direct generator-feature transfer as a separate control.

**Cost of the loop.** Against the corresponding no-repair run at an identical update budget, the loop raises wall-clock time by 29 to 46 percent across the three source regimes, most of it diffusion sampling. The cost grows with the anchor budget and the variants per anchor, and not with the number of cycles at fixed total budget. We have not optimized it: the sampler runs at its default step count, variants are never cached across cycles, and a distilled few-step generator would cut the dominant term directly.

**Alternative acquisition and repair pipelines.** Appendix Table 5 compares BRIDGE against TADA-style augmentation, an AIDE-style VLM pipeline, inverse-cluster frequency, DiffAug and direct SD3 feature distillation under one matched protocol at a single repair stage. The six span 1.67 points of four-task transfer, TADA-SSL at  $20.00 \pm 0.98$  and BRIDGE at  $18.88 \pm 0.58$ , so none separates from any other, as Section 6.4 predicts for one intervention.

**Sample selection.** Table 4 compares three upper-tail mode-window variants with top-tail and uniform sampling. All three are much stronger, and q50–99 and q1–99 stay close to the default, so the exact lower cutoff is not critical once selection is restricted to a sparse band and diversified within it. We use q75–99 and do not claim it is universally optimal.

Table 4: Selection and cycle ablations on ImageNet-100-LT. This matched ablation family is independent of the source-regime runs in Table 1. Values average all seven downstream tasks over three seeds. The nearby mode-window results show that the exact lower percentile is not critical. Uniform and top-tail selection test the dense-allocation and extreme-sparsity alternatives.

| Group     | Setting            | Linear probe   | $k$ NN         |
|-----------|--------------------|----------------|----------------|
| Selection | Mode-window q1–99  | $42.1 \pm 0.7$ | $33.4 \pm 0.3$ |
|           | Mode-window q50–99 | $42.2 \pm 0.9$ | $33.6 \pm 0.3$ |
|           | Mode-window q75–99 | $42.3 \pm 0.7$ | $33.6 \pm 0.5$ |
|           | Uniform            | $35.5 \pm 1.2$ | $28.4 \pm 0.6$ |
|           | Top-tail           | $34.3 \pm 1.2$ | $27.3 \pm 0.6$ |
| Schedule  | 2 cycles           | $35.0 \pm 1.4$ | $27.9 \pm 0.5$ |
|           | 5 cycles           | $42.3 \pm 0.7$ | $33.6 \pm 0.5$ |
|           | 10 cycles          | $34.1 \pm 1.7$ | $28.1 \pm 0.6$ |

**Reading the robustness sweep.** Appendix Table 20 varies the candidate-pool multiplier, upper cutoff, neighbourhood size, distance and selection strategy one at a time. Neighbourhood size is the parameter most likely to be dataset-dependent, and over  $k \in \{10, 25, 50, 100, 200\}$  the seven-task average moves between 26.80 and 27.84, so we take  $k = 100$  as a default and claim no optimum. The table carries a scale for its own resolution: cosine scoring rescales the normalized  $L_2$  score by a positive constant, so those two rows are the same experiment executed twice and differ by up to 1.3 points. Differences of that size elsewhere are not readable as effects of the knob varied. The contrast with the 6.6 to 8.0 point gap to uniform and top-tail acquisition identifies the sparse band as the component that carries the result.

**Cycle count.** Against 2 cycles, 5 cycles improve the seven-task averages by 7.3 points for linear probing and 5.7 for  $k$ NN at fixed total repair volume, and remain stronger than 10, so longer schedules do not help monotonically. Across backbones, ResNet-50 leads at 42.3 against 34.8 for ViT-B.

## 6.4 WHERE THE ACQUISITION RULE ACQUIRES ITS EFFECT

Figure 2 spends one identical acquisition budget on each decile-scale band of the  $k$ NN score distribution and changes nothing else in the loop. The nine seven-task averages lie between 26.95 and 28.27, a spread of 1.32 points, the same size as the separation between two robustness rows that are provably

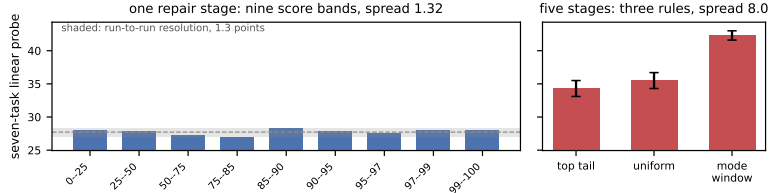


Figure 2: One repair stage against five. Left: nine bands of the  $k$ NN score distribution, each spending an identical budget after a single stage, all inside the shaded envelope, which is the run-to-run resolution of the pipeline measured by two rows of the robustness table that are provably the same experiment. Right: three acquisition rules under five stages at the same repair volume and optimizer budget. Rules indistinguishable after one intervention separate by 8.0 points once the signal is recomputed between them.

the identical experiment. One stage does not distinguish the densest quartile from the extreme tail, and the selector sweep behaves the same way.

Under five stages the same rules spread by 8.0 points. The protocols differ in how often the acquisition signal is recomputed, and not in the rules, the repair volume or the optimizer budget.

A single well-placed batch of 2,500 generated images is therefore worth no more than a badly placed one, and the value of a targeted rule accumulates over the loop: each intervention changes the representation that chooses the next, and small differences compound. Proposition 2 describes that behavior, and it separates BRIDGE from a one-shot rule with the same components. It also limits the theory, since the interior optimum of Proposition 1 is not detectable at one stage.

## 6.5 REPRESENTATION ANALYSIS

On a fixed panel of original images, MoCo v3 with ViT-S gains  $21.97 \pm 1.58$  effective rank,  $0.296 \pm 0.019$  spectral entropy and  $3.80 \pm 0.80$  1-NN accuracy, the first two surviving Holm correction. Local support also becomes more even: the Gini coefficient of the  $k$ NN radius over that panel for DINO falls on all three seeds against the unrepaired arm at the same optimizer stage,  $0.1235 \pm 0.0121$  to  $0.1089 \pm 0.0093$ , a paired change of  $-0.0147 \pm 0.0030$  at  $p = .014$ . Neither measure is a universal account: SimCLR improves transfer while its global effective rank falls.

## 7 SCOPE AND LIMITATIONS

BRIDGE depends on informative encoder neighborhoods and faithful local generation, and the frozen generator imports biases from its pretraining data. Its score band, neighborhood size and budget are operating choices and not universal constants. Four SSL objectives improve on 12 or 13 of 14 measurements each, and we have not shown that every objective benefits. The fixed-space analysis verifies where support is inserted, not how every learned neighborhood evolves. Corpus-scale effectiveness beyond our 10k web subsets remains open.

## 8 CONCLUSION

We formulate skewed-data SSL as a feedback problem in which the learner allocates a finite budget to change its future source distribution, and BRIDGE realizes those principles through a label-free loop that improves transfer more reliably than uniform addition, maximal-outlier generation, one-shot generation and the language-mediated alternatives we tested. Where the budget is spent means nothing for one intervention and 8.0 points once the representation choosing the target is itself repaired.

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## A COMPARISON WITH RELATED ACQUISITION METHODS

Table 6 makes the comparison axes explicit. Classical oversampling and TADA receive labels or supervised learning signals. DOPING uses latent rarity but targets anomaly detection and does not close the loop through a changing SSL representation. GenRep and StableRep make synthetic images or latent neighborhoods the source of contrastive views without allocating a repair budget over an observed skewed corpus. DiffAug is the closest iterative SSL antecedent, but it jointly learns a generator for generic positive views and does not ask where a frozen generator should intervene. AIDE and industrial data-engine systems establish the broader model–data feedback pattern, while using supervised task failures, VLM/LLM reasoning, annotation, and domain-specific verification. BRIDGE studies the label-free special case in which local support in the current SSL representation is both the diagnostic and acquisition signal. Our VLM rare-concept baseline adapts the AIDE idea by clustering frozen CLIP features, selecting inversely to cluster occupancy, captioning the selected images, and generating from text. The paired text-to-image and captioned SDEdit controls separately test whether retaining the selected image as a local condition matters.

The generator literature also identifies the origins of the repair operator and its controls. SDEdit supplies the image-editing mechanism, while synthetic-recognition studies establish the importance of prompts, filtering, diversity, and domain gap. DreamTeacher and DIFT motivate direct generator-feature transfer, and REPA studies the reverse direction from a visual encoder into a diffusion model. Omniverse Replicator provides non-archival systems context for iterative synthetic-data pipelines. BRIDGE brings these tools into a label-free local acquisition problem with a finite intervention budget and repeated re-estimation.

## B REPRESENTATION DIAGNOSTICS

The fixed-space intervention test freezes the cycle-0 encoder and queries the same original images before and after inserting all generated samples. First-cycle anchors are matched one-to-one within class to original images with the closest initial  $k$ NN radius that are never selected in any cycle.

Table 5: Empirical comparison with the closest acquisition and repair pipelines on ImageNet-100-LT, all under one matched protocol with a single repair stage, identical anchors budget and identical optimizer budget. Four-task transfer averages Cars, Aircraft, Flowers and Pets under a linear probe; representation 1-NN is measured on the source set. No pipeline separates from any other, BRIDGE included, which is the single-stage regime bounded by Proposition 3. The five-cycle comparisons in Section 6.4 are where acquisition rules do separate.

| Method                                  | Seeds | Four-task transfer | Representation 1-NN |
|-----------------------------------------|-------|--------------------|---------------------|
| Uniform + SD3                           | 3     | $18.33 \pm 0.32$   | $28.97 \pm 2.11$    |
| Top-tail + SD3                          | 3     | $19.51 \pm 0.22$   | $28.63 \pm 1.48$    |
| Cluster-frequency + SD3                 | 3     | $18.66 \pm 0.10$   | $28.66 \pm 2.42$    |
| TADA-SSL + SD3                          | 3     | $20.00 \pm 0.98$   | $28.78 \pm 2.22$    |
| BRIDGE + SD3                            | 3     | $18.88 \pm 0.58$   | $28.69 \pm 1.82$    |
| Mode-window + conventional augmentation | 3     | $18.52 \pm 0.65$   | $28.91 \pm 1.96$    |
| DiffAug adaptation                      | 3     | $11.50 \pm 1.87$   | $19.39 \pm 2.78$    |
| DiffAug adaptation + BRIDGE             | 3     | $10.42 \pm 2.17$   | $20.28 \pm 2.64$    |
| SD3 feature distillation                | 3     | $18.02 \pm 0.47$   | $26.27 \pm 2.29$    |
| SD3 feature distillation + BRIDGE       | 3     | $18.87 \pm 0.45$   | $29.45 \pm 2.26$    |

Table 6: Closest methods and the dimensions relevant to the acquisition problem. “Feedback” means that the learner or task model changes the next acquisition decision.

| Method                                                        | Labels or task loss         | Acquisition signal                 | Data action                     | Feedback | Primary goal            |
|---------------------------------------------------------------|-----------------------------|------------------------------------|---------------------------------|----------|-------------------------|
| SMOTE (Chawla et al., 2002)                                   | class labels                | class minority                     | interpolate examples            | no       | supervised imbalance    |
| DOPING (Lim et al., 2018)                                     | no labels                   | latent edge density                | latent oversampling             | no       | anomaly detection       |
| Concept pruning (Abbas et al., 2024)                          | no task labels              | cluster complexity                 | remove examples                 | no       | efficient pretraining   |
| DA-Fusion (Trabucco et al., 2024)                             | labels / few-shot set       | class and exemplar                 | image-conditioned diffusion     | no       | supervised recognition  |
| TADA (Nguyen et al., 2026)                                    | supervised loss dynamics    | slow-learned examples              | faithful diffusion variants     | no       | supervised recognition  |
| GenRep / StableRep (Jahanian et al., 2022; Tian et al., 2023) | prompts or generator latent | global sampling                    | synthetic SSL views             | no       | representation learning |
| DiffAug (Zang et al., 2024)                                   | no labels                   | learned positive-view objective    | jointly learned diffusion views | yes      | contrastive learning    |
| AIDE (Liang et al., 2024)                                     | task labels and failures    | VLM/LLM failure analysis           | curate, generate, annotate      | yes      | driving detection       |
| BRIDGE                                                        | no labels                   | local support in current SSL space | budgeted local variants         | yes      | source-support repair   |

Table 7 reports the direct anchor-versus-control comparison. The learned-space comparison retrain the encoder after the intervention and evaluates the same original-image panel. Its local effect is mixed for SimCLR, while the global response depends on the objective. We therefore treat the fixed-space result as evidence about where new support is inserted, not as evidence that all learned representations must change in the same geometric direction.

## C PROOFS FOR THE SUPPORT-ALLOCATION FRAMEWORK

**Proposition 4** (Sparse-region marginal value).  $\Delta_j(m_j)$  is positive and strictly decreases with  $n_j + m_j$ . When two regions have equal transfer mass and difficulty, the region with less effective support has the larger marginal value.

**Proposition 5** (Closed-form interior optimum). Let  $b(d) = b_0 d^\beta$  with  $b_0 > 0$  and  $\beta \in (0, 1]$ , and let  $\rho(d) = \rho_0 e^{-\gamma d}$  with  $\gamma > 0$ . Then  $g$  is strictly decreasing on  $(0, \infty)$  for every  $\lambda \geq 0$ , so Proposition 1 applies whenever  $g$  changes sign on the interval. If  $\lambda = 0$  the maximizer is  $d^* = \beta/\gamma$ , independent of  $b_0$  and  $\rho_0$ .

**Proof of Proposition ??.** Sort the distances from  $z$  to points in  $Z$  and denote their  $k$ th order statistic by  $r_k(z; Z)$ . Adding  $A$  preserves every old candidate distance and may introduce smaller ones. The  $k$ th order statistic can therefore only decrease. It decreases strictly when an added distance enters the first  $k$  positions and displaces a strictly larger distance.

**Proof of Proposition 4.** Let  $x = n_j + m_j + \kappa$ . Since  $f(x) = x^{-\gamma}$  is positive, strictly decreasing, and strictly convex for  $x > 0$ , the forward difference  $f(x) - f(x+1)$  is positive and strictly decreases with  $x$ . Multiplication by  $p_j a_j > 0$  preserves both properties. Equal  $p_j a_j$  therefore leaves effective support as the sole determinant of the marginal ordering.

Table 7: Local support added in the frozen cycle-0 representation, reported as mean  $\pm$  standard deviation over three seeds. Controls are matched one-to-one within class on initial  $k$ NN radius and are never selected. Negative radius DiD and positive purity DiD favor treated anchors.

|               | $\Delta r$ anchor      | $\Delta r$ control     | Radius DiD             | Purity DiD           |
|---------------|------------------------|------------------------|------------------------|----------------------|
| Mean $\pm$ SD | $-0.00273 \pm 0.00062$ | $-0.00139 \pm 0.00004$ | $-0.00134 \pm 0.00067$ | $+0.0134 \pm 0.0032$ |

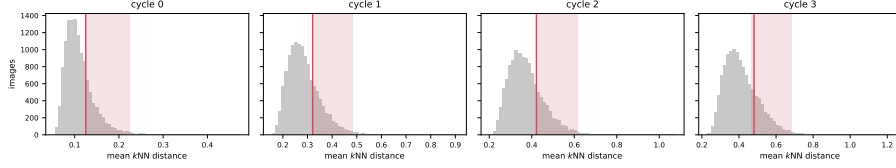


Figure 3: Score distribution over repair cycles for the same run, with the q75–q99 acquisition band shaded and the modal bin marked. The distribution stays right-skewed while its scale grows, so the band tracks a moving target and the selector re-derives it from the current representation at every cycle.

**Proof of Proposition 1.** Rewrite the utility as  $u(d) = \rho(d)(b(d) + \lambda) - \lambda$ . Its derivative is

$$u'(d) = \rho(d)(b(d) + \lambda) \left[ \frac{b'(d)}{b(d) + \lambda} + \frac{\rho'(d)}{\rho(d)} \right] = \rho(d)(b(d) + \lambda)g(d).$$

The factor preceding  $g(d)$  is positive. Since  $g$  is strictly decreasing and changes sign across the interval, it has one root  $d^*$  by continuity. Hence  $u$  increases before  $d^*$  and decreases after it, making  $d^*$  the unique global maximizer.

**Proof of Proposition 5.** With  $b(d) = b_0 d^\beta$  and  $\rho(d) = \rho_0 e^{-\gamma d}$  we have  $\rho'(d)/\rho(d) = -\gamma$ , so

$$g(d) = \frac{b_0 \beta d^{\beta-1}}{b_0 d^\beta + \lambda} - \gamma = \frac{\beta d^{\beta-1}}{d^\beta + c} - \gamma, \quad c = \lambda/b_0 \geq 0.$$

Write  $h(d) = \beta d^{\beta-1}(d^\beta + c)^{-1}$ . Differentiating,

$$h'(d) = \frac{\beta d^{\beta-2}[(\beta-1)(d^\beta + c) - \beta d^\beta]}{(d^\beta + c)^2} = \frac{\beta d^{\beta-2}[(\beta-1)c - d^\beta]}{(d^\beta + c)^2}.$$

For  $\beta \in (0, 1]$  we have  $(\beta-1)c \leq 0 < d^\beta$  for all  $d > 0$ , so the bracket is strictly negative and  $h'(d) < 0$ . Hence  $g = h - \gamma$  is strictly decreasing for every  $\lambda \geq 0$ . When  $\lambda = 0$  we have  $c = 0$  and  $g(d) = \beta/d - \gamma$ , whose unique root is  $d^* = \beta/\gamma$ ;  $g$  is positive below it and negative above it, so  $d^*$  is the maximizer by the argument of Proposition 1.

**Proof of Proposition 3.** Write  $f(x) = x^{-\gamma}$ . For  $x > 0$  the mean value theorem gives  $f(x) - f(x+1) = \gamma \xi^{-\gamma-1}$  for some  $\xi \in (x, x+1)$ , so

$$0 < \Delta_j(m_j) = p_j a_j [f(n_j + m_j + \kappa) - f(n_j + m_j + 1 + \kappa)] \leq p_j a_j \gamma (n_j + \kappa)^{-\gamma-1},$$

using  $\xi > n_j + m_j + \kappa \geq n_j + \kappa$  and  $m_j \geq 0$ . Both  $\mathcal{B}(\mathbf{m})$  and  $\mathcal{B}(\mathbf{m}')$  are obtained from the common baseline  $\mathcal{B}(\mathbf{0})$  by subtracting sums of at most  $B$  such marginals, since  $\sum_j m_j = \sum_j m'_j = B$ . Each sum therefore lies in  $[0, B\gamma \max_j p_j a_j (n_j + \kappa)^{-\gamma-1}]$ , and the difference of two numbers drawn from an interval is bounded by its length. The stated bound follows, and monotonicity of  $(n + \kappa)^{-\gamma-1}$  in  $n$  gives the  $n_{\min}$  form.

**Proof of Proposition 2.** For region  $j$ , define the marginal sequence  $\delta_{j,t} = U_j(n_j + t) - U_j(n_j + t - 1)$  for  $t \geq 1$ . Discrete concavity makes each sequence nonincreasing. Any feasible allocation of  $B$  repairs selects a prefix of length  $m_j$  from each sequence. Greedy allocation repeatedly chooses the largest available next element. After  $B$  steps it has selected the  $B$  largest prefix-feasible marginals. If another allocation had larger utility, it would contain a selected marginal smaller than an available unselected marginal, and exchanging them would improve that allocation. This contradicts optimality. A frozen allocation can fail this rule because it preserves the initial ordering after selected regions move to smaller marginals.

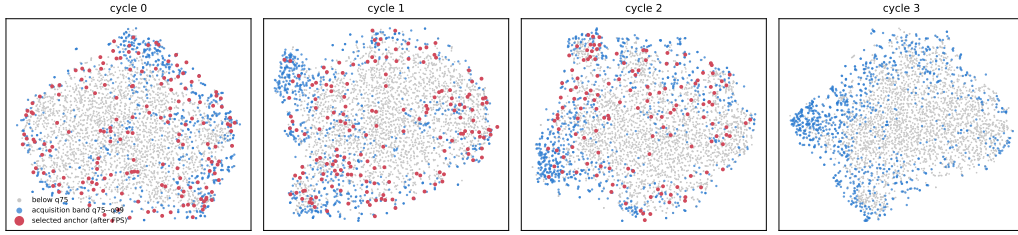


Figure 4: Where the acquisition budget is spent, projected to two dimensions with t-SNE, for one ImageNet-100-LT run. Grey marks scores below the 75th percentile, blue the q75–q99 acquisition band, and red the anchors farthest-point sampling selects from that band. The band occupies the periphery of the projection: over the panel shown, the rank correlation between  $k$ NN score and distance from the projection centre is +0.51, and band members sit at mean radius 34.6 against 25.5 for the rest. The anchors are spread across the band and not clustered inside it, which is the behaviour farthest-point sampling is included to produce. Each panel is projected separately because the encoder changes between cycles, so positions are comparable within a panel and not across panels. The panel contains original images only, so it shows which regions are targeted and not what the inserted samples do to them; Table 7 measures the latter directly.

## D IMPLEMENTATION AND REPRODUCIBILITY

**A corrected DINO recipe.** The DINO arm was rebuilt after we found that our earlier implementation omitted the bottleneck projection head and applied solarization to the wrong crops. Its baseline is now competitive with SimCLR and MoCo v3 on the same schedule, and the earlier report of DINO as a null came from encoders published before the fault described next.

**A checkpointing fault and the reruns it forced.** Each cycle constructs a fresh trainer whose step counter restarts at zero and stops at the same step limit, and the checkpoint callback, reused across cycles, treats a save at a step it has already written as a duplicate and skips it. Runs configured with a per-cycle step cap therefore published their first-cycle encoder however far the later cycles diverged, which understates any method whose benefit accumulates over cycles. We found this when two policy conditions reproduced each other to the last bit while their anchor manifests overlapped on 14 of 500 anchors from the second cycle onward. Every paired ViT-S cell in Table 3 was rerun with the callback state reset between cycles, 19 runs and 416 wall-clock hours, and we verify before publication that no two conditions in a table share an encoder fingerprint. Runs whose cycles end on an epoch boundary rather than a step cap were never affected.

**Core training configuration.** The main experiments use ResNet-50 encoders. Most baseline, BRIDGE, and source-regime comparisons use 5 cycles with 100 epochs per cycle. The default BRIDGE configuration uses mean  $k$ NN distance with  $k = 100$ , selects 500 samples per cycle, and generates 5 images per selected sample. The default optimizer is AdamW with weight decay  $10^{-4}$ , and method-specific learning rates and temperatures follow the settings used in each experiment.

**OOD scoring and selection.** The submitted experiments used raw squared  $\ell_2$  distance for OOD scoring and normalized features for farthest-point sampling. Because raw feature norms can confound cross-objective comparisons, the new controlled experiments use normalized  $\ell_2$  throughout and report raw  $\ell_2$ , normalized  $\ell_2$ , cosine, and multiscale- $k$  variants as a robustness study. The mode-window selector restricts the OOD distribution to the 75th to 99th percentile range, builds a histogram with the code-level auto-bin heuristic, forms a mode-centered candidate pool of size  $4B$ , and then applies greedy farthest-point sampling to select the final  $B$  anchors.

**Generators.** The main method uses Stable Diffusion 3 in image-to-image mode. The generation ablation replaces it with FLUX or with oracle re-population, which adds the withheld real images and runs no generator at all. In the PASS and DiffusionDB experiments, we increase the SD3 batch size to 12 to fit the 24-hour wall-time limit.

**Source data specifics.** PASS and DiffusionDB use `train[:10000]` with pretraining split override `[1.0, 0.0, 0.0]`. This keeps the source data budget aligned across methods while avoiding unnecessary pretraining validation overhead. The long-tailed ImageNet-100, CIFAR-10, and CIFAR-100 source regimes use their standard long-tailed source configurations, without the 10k subset restriction used for PASS and DiffusionDB.

**Seeds and reporting.** All main claims are supported by three seed averages. Tables report mean  $\pm$  standard deviation over those seeds. The exact table values are computed directly from the three runs corresponding to each reported setting.

**Code availability.** An anonymous code release accompanying this submission is available at <https://anonymous.4open.science/status/FOMO-0488>.

## E COMPUTE RESOURCES AND RUNTIME

Most reported experiments use one node with two NVIDIA A100 80GB GPUs and a 24-hour wall-time limit. The DINO arm uses the reference multi-crop setting of two global crops and eight local crops at  $96 \times 96$ , which costs about 3.5 global-crop forward passes per image and fits on a single 48GB card at the batch size used here. The 454 completed runs retained on disk for this paper total approximately 5,471 wall-clock hours. Runs on the two-GPU nodes consume roughly twice that in GPU hours, while the paired ViT-S reruns hold a single card each. The 19 reruns that replaced encoders affected by the checkpointing fault account for 416 hours, and the baselines and controls that appear only in the appendix account for a further 2,987. The figure excludes failed runs, timeout runs, cache migration jobs and pilot runs, so the project consumed more than the successful totals alone. This excludes failed runs, timeout runs, cache migration jobs, and other debugging or pilot jobs, so the full project required more compute than the successful totals alone.

Representative mean wall-clock times over three seeds are as follows. On ImageNet-100-LT source pretraining, SimCLR takes 16.9 hours and BRIDGE takes 21.8 hours. On PASS, SimCLR takes 10.2 hours and BRIDGE takes 14.9 hours. On DiffusionDB, SimCLR takes 11.6 hours and BRIDGE takes 16.4 hours. The repair loop therefore raises wall-clock time by 29 to 46 percent over the corresponding no-repair run, of which the embedding pass accounts for a small fraction and diffusion sampling for the remainder.

## F ASSETS AND TERMS OF USE

We use existing datasets and pretrained generators under their original release terms.

**Source datasets.** ImageNet-100-LT is derived from ImageNet, whose official access terms restrict the images to non commercial research and educational use. PASS is released under Creative Commons Attribution 4.0. DiffusionDB is released under CC0-1.0. CIFAR-10, CIFAR-100, Stanford Cars, FGVC Aircraft, Oxford Flowers, and Oxford-IIIT Pets are used from their official academic benchmark releases. When an official page provides download terms instead of a standard software style license, we follow those original terms.

**Generators.** Stable Diffusion 3 Medium is used under the Stability AI Community License. FLUX.1-dev is used only in the ablation and is subject to the FLUX.1-dev Non-Commercial License. We do not redistribute model weights, third party datasets, or generated image corpora as part of this submission.

## G BROADER IMPACT

A method that improves representation quality under biased source data can reduce the need for manual relabeling and can make SSL more useful in domains where balanced data are difficult to collect. That is a positive outcome. At the same time, our method relies on large image generators and on web-sourced datasets. These carry familiar risks related to license compliance, privacy, and

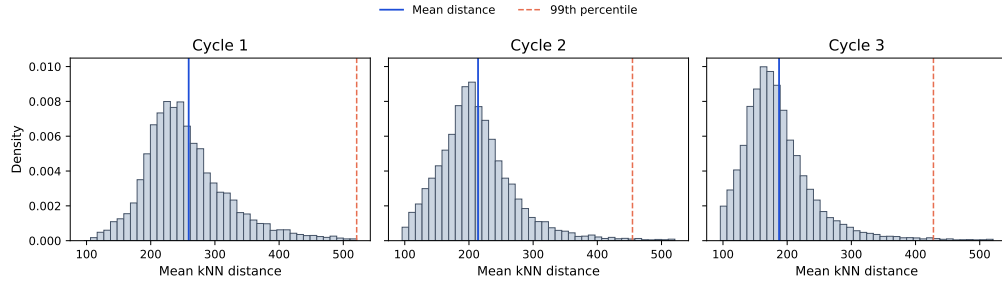


Figure 5: Representative OOD histograms across BRIDGE cycles from an ImageNet-100-LT run. Solid blue lines mark the mean distance and dashed orange lines mark the 99th percentile. The acquisition distribution and its modal shoulder change as training and repair progress.

unintended memorization or harmful content. Our experiments stay within existing dataset and model release terms and do not release new generators or scraped image collections. Any future deployment would still require a task-specific review of fairness, privacy, and content safety.

## H FULL RESULTS

**Where the temperature-schedule hybrid helps.** Combining BRIDGE with TS lowers the seven-task average in every source regime, and it does win 7 of the 70 individual cells. Six of those seven fall on CIFAR-10 and ImageNet-100-LT, the two long-tailed downstream targets, and the largest are +3.3 and +2.8 points on ImageNet-100-LT and CIFAR-10 under a PASS and a CIFAR-10-LT source. The fine-grained targets, Cars, Aircraft, Flowers and Pets, lose in almost every case. TS reweights the contrastive temperature toward rare samples, which sharpens the head-to-tail boundary the long-tailed targets are scored on, while BRIDGE adds support that broadens transfer. We read the hybrid as two interventions serving different targets and not as an interaction, and we report the components separately for that reason.

Table 8: Full linear-probe transfer results across all source datasets and comparison methods.

| Source          | Method              | C10                              | C100                             | Cars                             | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|-----------------|---------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| ImageNet-100-LT | SimCLR              | 73.9 $\pm$ 0.5                   | 46.8 $\pm$ 0.4                   | 13.7 $\pm$ 0.3                   | 19.6 $\pm$ 0.2                   | 35.9 $\pm$ 7.2                   | 38.6 $\pm$ 1.5                   | 50.4 $\pm$ 0.5                   | 39.8 $\pm$ 1.5                   |
|                 | TS                  | 72.5 $\pm$ 0.4                   | 44.0 $\pm$ 0.5                   | 12.5 $\pm$ 1.2                   | 15.0 $\pm$ 0.1                   | 35.7 $\pm$ 1.8                   | 40.6 $\pm$ 2.2                   | <b>56.7 <math>\pm</math> 1.2</b> | 39.6 $\pm$ 1.1                   |
|                 | SDCLR               | 41.1 $\pm$ 1.7                   | 18.0 $\pm$ 0.6                   | 1.4 $\pm$ 0.3                    | 2.3 $\pm$ 0.5                    | 10.8 $\pm$ 1.7                   | 10.4 $\pm$ 0.3                   | 21.9 $\pm$ 0.6                   | 15.1 $\pm$ 0.8                   |
|                 | BRIDGE (ours)       | <b>75.8 <math>\pm</math> 0.2</b> | <b>49.3 <math>\pm</math> 0.2</b> | <b>16.4 <math>\pm</math> 1.2</b> | <b>22.4 <math>\pm</math> 0.4</b> | <b>36.8 <math>\pm</math> 1.9</b> | <b>42.9 <math>\pm</math> 1.0</b> | 53.8 $\pm$ 0.1                   | <b>42.5 <math>\pm</math> 0.7</b> |
|                 | BRIDGE+TS (ours)    | 74.2 $\pm$ 0.5                   | 45.6 $\pm$ 0.8                   | 9.4 $\pm$ 0.5                    | 15.0 $\pm$ 0.2                   | 25.5 $\pm$ 2.5                   | 33.1 $\pm$ 2.8                   | 48.3 $\pm$ 0.4                   | 35.9 $\pm$ 1.1                   |
|                 | BRIDGE+SDCLR (ours) | 43.8 $\pm$ 1.7                   | 18.4 $\pm$ 0.7                   | 2.0 $\pm$ 0.3                    | 2.5 $\pm$ 0.8                    | 10.0 $\pm$ 0.8                   | 12.2 $\pm$ 3.1                   | 21.8 $\pm$ 0.5                   | 15.8 $\pm$ 1.1                   |
| CIFAR-10-LT     | SimCLR              | 67.8 $\pm$ 0.2                   | 38.0 $\pm$ 1.0                   | 6.5 $\pm$ 0.4                    | 16.0 $\pm$ 0.9                   | <b>19.2 <math>\pm</math> 1.6</b> | 24.0 $\pm$ 1.3                   | 32.0 $\pm$ 0.7                   | 29.1 $\pm$ 0.9                   |
|                 | TS                  | 68.5 $\pm$ 1.3                   | 36.0 $\pm$ 1.7                   | 4.3 $\pm$ 0.2                    | 10.3 $\pm$ 0.1                   | 13.1 $\pm$ 2.0                   | 19.7 $\pm$ 2.3                   | 31.5 $\pm$ 1.2                   | 26.2 $\pm$ 1.3                   |
|                 | SDCLR               | 39.6 $\pm$ 1.1                   | 15.6 $\pm$ 0.2                   | 1.1 $\pm$ 0.1                    | 2.4 $\pm$ 0.9                    | 12.0 $\pm$ 1.6                   | 9.1 $\pm$ 1.8                    | 16.5 $\pm$ 1.6                   | 13.8 $\pm$ 1.1                   |
|                 | BRIDGE (ours)       | 78.1 $\pm$ 0.3                   | <b>49.8 <math>\pm</math> 0.4</b> | <b>9.2 <math>\pm</math> 0.8</b>  | <b>17.7 <math>\pm</math> 1.0</b> | 18.2 $\pm$ 1.1                   | <b>27.4 <math>\pm</math> 2.7</b> | 39.0 $\pm$ 0.6                   | <b>34.2 <math>\pm</math> 1.0</b> |
|                 | BRIDGE+TS (ours)    | <b>78.6 <math>\pm</math> 0.3</b> | 46.1 $\pm$ 0.3                   | 7.5 $\pm$ 0.9                    | 11.1 $\pm$ 0.7                   | 14.5 $\pm$ 0.6                   | 25.1 $\pm$ 2.7                   | <b>39.6 <math>\pm</math> 0.8</b> | 31.8 $\pm$ 0.9                   |
|                 | BRIDGE+SDCLR (ours) | 40.9 $\pm$ 0.7                   | 17.9 $\pm$ 0.5                   | 1.4 $\pm$ 0.1                    | 2.2 $\pm$ 0.6                    | 11.0 $\pm$ 2.4                   | 10.0 $\pm$ 2.7                   | 21.0 $\pm$ 0.6                   | 14.9 $\pm$ 1.1                   |
| CIFAR-100-LT    | SimCLR              | 59.0 $\pm$ 1.5                   | 31.1 $\pm$ 1.6                   | 4.1 $\pm$ 0.3                    | 11.2 $\pm$ 1.0                   | 12.6 $\pm$ 3.4                   | 20.1 $\pm$ 1.0                   | 28.7 $\pm$ 1.3                   | 23.8 $\pm$ 1.4                   |
|                 | TS                  | 56.3 $\pm$ 0.2                   | 28.0 $\pm$ 0.1                   | 2.8 $\pm$ 0.6                    | 8.5 $\pm$ 0.2                    | 15.5 $\pm$ 2.5                   | 16.2 $\pm$ 3.1                   | 26.2 $\pm$ 1.8                   | 21.9 $\pm$ 1.2                   |
|                 | SDCLR               | 37.1 $\pm$ 0.8                   | 15.7 $\pm$ 0.2                   | 1.9 $\pm$ 0.1                    | 2.4 $\pm$ 0.7                    | 9.4 $\pm$ 1.0                    | 9.9 $\pm$ 1.7                    | 16.2 $\pm$ 0.3                   | 13.2 $\pm$ 0.7                   |
|                 | BRIDGE (ours)       | <b>74.2 <math>\pm</math> 0.5</b> | <b>48.1 <math>\pm</math> 0.4</b> | <b>8.3 <math>\pm</math> 1.0</b>  | <b>18.2 <math>\pm</math> 0.7</b> | <b>24.4 <math>\pm</math> 2.3</b> | 27.6 $\pm$ 0.7                   | 39.8 $\pm$ 1.1                   | <b>34.4 <math>\pm</math> 1.0</b> |
|                 | BRIDGE+TS (ours)    | 73.7 $\pm$ 0.1                   | 47.3 $\pm$ 0.5                   | 7.9 $\pm$ 0.6                    | 14.1 $\pm$ 0.5                   | 18.2 $\pm$ 3.2                   | <b>28.4 <math>\pm</math> 0.4</b> | <b>41.3 <math>\pm</math> 0.3</b> | 33.0 $\pm$ 0.8                   |
|                 | BRIDGE+SDCLR (ours) | 42.0 $\pm$ 1.5                   | 16.9 $\pm$ 0.9                   | 1.3 $\pm$ 0.1                    | 2.2 $\pm$ 0.6                    | 10.6 $\pm$ 0.8                   | 8.3 $\pm$ 0.8                    | 20.2 $\pm$ 0.9                   | 14.5 $\pm$ 0.8                   |
| PASS-10k        | SimCLR              | 72.3 $\pm$ 0.2                   | 45.5 $\pm$ 0.5                   | 11.0 $\pm$ 0.2                   | 17.5 $\pm$ 1.4                   | 33.3 $\pm$ 1.3                   | 33.3 $\pm$ 0.9                   | 44.7 $\pm$ 0.9                   | 36.8 $\pm$ 0.8                   |
|                 | TS                  | 71.5 $\pm$ 0.2                   | 43.5 $\pm$ 0.4                   | 10.2 $\pm$ 1.2                   | 17.2 $\pm$ 0.2                   | 33.1 $\pm$ 2.1                   | 36.4 $\pm$ 1.8                   | 49.4 $\pm$ 0.4                   | 37.3 $\pm$ 0.9                   |
|                 | SDCLR               | 42.2 $\pm$ 1.1                   | 17.7 $\pm$ 0.8                   | 1.8 $\pm$ 0.3                    | 2.2 $\pm$ 0.8                    | 10.6 $\pm$ 1.3                   | 8.2 $\pm$ 1.6                    | 20.8 $\pm$ 0.4                   | 14.8 $\pm$ 0.9                   |
|                 | BRIDGE (ours)       | <b>73.9 <math>\pm</math> 0.3</b> | <b>47.1 <math>\pm</math> 0.4</b> | <b>13.6 <math>\pm</math> 0.7</b> | <b>20.5 <math>\pm</math> 0.7</b> | <b>37.7 <math>\pm</math> 4.0</b> | <b>37.1 <math>\pm</math> 0.7</b> | 49.1 $\pm$ 0.5                   | <b>39.9 <math>\pm</math> 1.0</b> |
|                 | BRIDGE+TS (ours)    | 68.9 $\pm$ 0.6                   | 40.6 $\pm$ 0.7                   | 12.1 $\pm$ 0.4                   | 14.1 $\pm$ 0.5                   | 34.6 $\pm$ 2.9                   | 31.6 $\pm$ 1.5                   | <b>52.4 <math>\pm</math> 0.6</b> | 36.4 $\pm$ 1.0                   |
|                 | BRIDGE+SDCLR (ours) | 44.0 $\pm$ 0.9                   | 18.8 $\pm$ 0.9                   | 1.4 $\pm$ 0.3                    | 1.8 $\pm$ 0.9                    | 12.6 $\pm$ 2.6                   | 12.4 $\pm$ 1.3                   | 21.2 $\pm$ 1.1                   | 16.0 $\pm$ 1.1                   |
| DiffusionDB-10k | SimCLR              | 74.4 $\pm$ 0.5                   | 48.6 $\pm$ 0.2                   | 10.9 $\pm$ 0.7                   | 17.6 $\pm$ 0.8                   | 31.2 $\pm$ 2.4                   | 36.1 $\pm$ 1.3                   | 45.3 $\pm$ 0.5                   | 37.7 $\pm$ 0.9                   |
|                 | TS                  | 72.3 $\pm$ 0.3                   | 44.4 $\pm$ 0.3                   | 10.7 $\pm$ 0.6                   | 15.4 $\pm$ 0.6                   | 33.1 $\pm$ 1.1                   | <b>39.9 <math>\pm</math> 2.7</b> | 48.8 $\pm$ 0.2                   | 37.8 $\pm$ 0.8                   |
|                 | SDCLR               | 42.4 $\pm$ 0.2                   | 18.4 $\pm$ 0.2                   | 1.1 $\pm$ 0.2                    | 2.4 $\pm$ 0.1                    | 13.0 $\pm$ 3.7                   | 8.2 $\pm$ 1.4                    | 21.0 $\pm$ 0.6                   | 15.2 $\pm$ 0.9                   |
|                 | BRIDGE (ours)       | <b>75.9 <math>\pm</math> 0.4</b> | <b>49.9 <math>\pm</math> 0.5</b> | <b>12.8 <math>\pm</math> 1.0</b> | <b>20.0 <math>\pm</math> 0.4</b> | <b>35.7 <math>\pm</math> 1.1</b> | 37.7 $\pm$ 0.8                   | 49.1 $\pm$ 0.5                   | <b>40.1 <math>\pm</math> 0.7</b> |
|                 | BRIDGE+TS (ours)    | 67.8 $\pm$ 0.6                   | 39.1 $\pm$ 0.5                   | 9.4 $\pm$ 1.2                    | 12.4 $\pm$ 0.4                   | 30.3 $\pm$ 3.0                   | 33.7 $\pm$ 1.9                   | <b>50.8 <math>\pm</math> 0.6</b> | 34.8 $\pm$ 1.2                   |
|                 | BRIDGE+SDCLR (ours) | 43.8 $\pm$ 0.9                   | 18.5 $\pm$ 0.3                   | 1.3 $\pm$ 0.4                    | 2.5 $\pm$ 0.0                    | 14.3 $\pm$ 1.6                   | 11.4 $\pm$ 0.6                   | 21.9 $\pm$ 0.6                   | 16.2 $\pm$ 0.6                   |

Table 9: Full  $k$ NN transfer results across all source datasets and comparison methods.

| Source          | Method              | C10                              | C100                             | Cars                            | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|-----------------|---------------------|----------------------------------|----------------------------------|---------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| ImageNet-100-LT | SimCLR              | 68.6 $\pm$ 0.4                   | 37.3 $\pm$ 0.2                   | 6.0 $\pm$ 0.7                   | 11.2 $\pm$ 0.4                   | 32.2 $\pm$ 0.7                   | 23.6 $\pm$ 0.3                   | 43.9 $\pm$ 0.9                   | 31.8 $\pm$ 0.5                   |
|                 | TS                  | 68.4 $\pm$ 0.3                   | 36.5 $\pm$ 0.1                   | 4.7 $\pm$ 0.2                   | 8.3 $\pm$ 0.5                    | 26.6 $\pm$ 1.0                   | 23.6 $\pm$ 1.7                   | <b>52.2 <math>\pm</math> 0.5</b> | 31.5 $\pm$ 0.6                   |
|                 | SDCLR               | 32.5 $\pm$ 0.4                   | 10.9 $\pm$ 0.2                   | 1.0 $\pm$ 0.3                   | 2.1 $\pm$ 0.1                    | 10.7 $\pm$ 0.7                   | 4.7 $\pm$ 0.2                    | 11.2 $\pm$ 0.2                   | 10.4 $\pm$ 0.3                   |
|                 | BRIDGE (ours)       | <b>69.0 <math>\pm</math> 0.5</b> | <b>37.8 <math>\pm</math> 0.3</b> | <b>7.0 <math>\pm</math> 0.2</b> | <b>11.7 <math>\pm</math> 0.2</b> | <b>36.5 <math>\pm</math> 0.4</b> | <b>25.3 <math>\pm</math> 1.0</b> | 48.1 $\pm$ 0.6                   | <b>33.6 <math>\pm</math> 0.4</b> |
|                 | BRIDGE+TS (ours)    | 65.4 $\pm$ 0.2                   | 33.6 $\pm$ 0.3                   | 3.1 $\pm$ 0.1                   | 7.1 $\pm$ 0.3                    | 18.2 $\pm$ 0.2                   | 15.6 $\pm$ 0.7                   | 40.8 $\pm$ 0.5                   | 26.3 $\pm$ 0.3                   |
|                 | BRIDGE+SDCLR (ours) | 36.3 $\pm$ 0.8                   | 13.0 $\pm$ 0.5                   | 1.1 $\pm$ 0.3                   | 2.4 $\pm$ 0.2                    | 9.7 $\pm$ 0.3                    | 5.4 $\pm$ 0.4                    | 13.9 $\pm$ 1.0                   | 11.7 $\pm$ 0.5                   |
| CIFAR-10-LT     | SimCLR              | 66.9 $\pm$ 0.4                   | 33.4 $\pm$ 0.3                   | 3.7 $\pm$ 0.4                   | 9.8 $\pm$ 0.7                    | 16.5 $\pm$ 0.8                   | 13.9 $\pm$ 0.8                   | 24.0 $\pm$ 0.8                   | 24.0 $\pm$ 0.6                   |
|                 | TS                  | 66.3 $\pm$ 0.6                   | 27.8 $\pm$ 0.4                   | 2.4 $\pm$ 0.6                   | 6.4 $\pm$ 0.2                    | 10.5 $\pm$ 0.7                   | 11.1 $\pm$ 0.4                   | 22.0 $\pm$ 0.9                   | 20.9 $\pm$ 0.5                   |
|                 | SDCLR               | 30.1 $\pm$ 0.3                   | 10.0 $\pm$ 0.3                   | 1.7 $\pm$ 0.3                   | 2.2 $\pm$ 0.2                    | 10.1 $\pm$ 0.7                   | 4.4 $\pm$ 0.4                    | 9.8 $\pm$ 0.3                    | 9.7 $\pm$ 0.4                    |
|                 | BRIDGE (ours)       | 74.8 $\pm$ 0.4                   | <b>41.0 <math>\pm</math> 0.4</b> | <b>4.8 <math>\pm</math> 0.5</b> | <b>10.0 <math>\pm</math> 0.2</b> | <b>24.2 <math>\pm</math> 0.6</b> | <b>16.2 <math>\pm</math> 0.7</b> | <b>29.7 <math>\pm</math> 0.3</b> | <b>28.7 <math>\pm</math> 0.4</b> |
|                 | BRIDGE+TS (ours)    | <b>77.6 <math>\pm</math> 0.7</b> | 35.3 $\pm$ 0.5                   | 3.1 $\pm$ 0.3                   | 6.6 $\pm$ 0.4                    | 16.3 $\pm$ 0.6                   | 12.5 $\pm$ 1.0                   | 26.7 $\pm$ 0.1                   | 25.5 $\pm$ 0.5                   |
|                 | BRIDGE+SDCLR (ours) | 34.4 $\pm$ 1.2                   | 11.4 $\pm$ 0.6                   | 1.2 $\pm$ 0.2                   | 2.2 $\pm$ 0.2                    | 8.8 $\pm$ 1.5                    | 5.0 $\pm$ 0.2                    | 11.7 $\pm$ 1.1                   | 10.7 $\pm$ 0.7                   |
| CIFAR-100-LT    | SimCLR              | 59.3 $\pm$ 0.8                   | 30.1 $\pm$ 0.9                   | 3.0 $\pm$ 0.3                   | 7.4 $\pm$ 0.4                    | 13.7 $\pm$ 0.2                   | 10.8 $\pm$ 0.4                   | 21.8 $\pm$ 1.1                   | 20.9 $\pm$ 0.6                   |
|                 | TS                  | 53.1 $\pm$ 0.6                   | 24.6 $\pm$ 0.4                   | 2.4 $\pm$ 0.4                   | 5.5 $\pm$ 0.3                    | 9.3 $\pm$ 0.4                    | 8.3 $\pm$ 0.2                    | 18.7 $\pm$ 1.0                   | 17.4 $\pm$ 0.5                   |
|                 | SDCLR               | 25.7 $\pm$ 1.2                   | 8.5 $\pm$ 0.6                    | 0.9 $\pm$ 0.4                   | 1.9 $\pm$ 0.2                    | 7.3 $\pm$ 0.3                    | 3.9 $\pm$ 0.3                    | 8.3 $\pm$ 1.1                    | 8.1 $\pm$ 0.6                    |
|                 | BRIDGE (ours)       | <b>69.3 <math>\pm</math> 0.5</b> | <b>39.5 <math>\pm</math> 0.1</b> | <b>4.8 <math>\pm</math> 0.6</b> | <b>8.9 <math>\pm</math> 0.1</b>  | <b>21.9 <math>\pm</math> 0.1</b> | <b>15.5 <math>\pm</math> 0.1</b> | <b>29.1 <math>\pm</math> 0.7</b> | <b>27.0 <math>\pm</math> 0.3</b> |
|                 | BRIDGE+TS (ours)    | 64.3 $\pm$ 0.4                   | 36.6 $\pm$ 0.4                   | 2.9 $\pm$ 0.3                   | 6.3 $\pm$ 0.1                    | 16.4 $\pm$ 0.5                   | 12.3 $\pm$ 0.0                   | 28.2 $\pm$ 0.6                   | 23.9 $\pm$ 0.3                   |
|                 | BRIDGE+SDCLR (ours) | 31.9 $\pm$ 0.4                   | 10.4 $\pm$ 0.5                   | 0.8 $\pm$ 0.1                   | 2.1 $\pm$ 0.5                    | 7.5 $\pm$ 1.6                    | 4.5 $\pm$ 0.1                    | 9.1 $\pm$ 0.7                    | 9.5 $\pm$ 0.6                    |
| PASS-10k        | SimCLR              | 67.2 $\pm$ 0.2                   | <b>37.0 <math>\pm</math> 0.3</b> | 5.4 $\pm$ 0.3                   | 10.3 $\pm$ 0.5                   | 30.4 $\pm$ 0.0                   | 16.7 $\pm$ 0.5                   | 38.3 $\pm$ 0.3                   | 29.3 $\pm$ 0.3                   |
|                 | TS                  | 65.2 $\pm$ 0.7                   | 34.6 $\pm$ 0.3                   | 4.4 $\pm$ 0.2                   | 7.9 $\pm$ 0.2                    | 24.5 $\pm$ 0.1                   | 14.4 $\pm$ 0.4                   | 39.3 $\pm$ 0.4                   | 27.2 $\pm$ 0.3                   |
|                 | SDCLR               | 32.4 $\pm$ 1.5                   | 11.5 $\pm$ 0.8                   | 1.3 $\pm$ 0.2                   | 2.0 $\pm$ 0.1                    | 8.6 $\pm$ 0.7                    | 5.2 $\pm$ 0.2                    | 11.1 $\pm$ 0.8                   | 10.3 $\pm$ 0.6                   |
|                 | BRIDGE (ours)       | <b>67.5 <math>\pm</math> 0.3</b> | 36.9 $\pm$ 0.3                   | <b>5.8 <math>\pm</math> 0.5</b> | <b>10.7 <math>\pm</math> 0.4</b> | <b>33.7 <math>\pm</math> 0.2</b> | <b>17.8 <math>\pm</math> 0.6</b> | <b>41.4 <math>\pm</math> 0.8</b> | <b>30.6 <math>\pm</math> 0.4</b> |
|                 | BRIDGE+TS (ours)    | 58.8 $\pm$ 0.5                   | 27.9 $\pm$ 0.5                   | 3.0 $\pm$ 0.2                   | 6.6 $\pm$ 0.1                    | 23.0 $\pm$ 0.3                   | 13.4 $\pm$ 0.4                   | 38.8 $\pm$ 0.8                   | 24.5 $\pm$ 0.4                   |
|                 | BRIDGE+SDCLR (ours) | 37.1 $\pm$ 2.0                   | 13.6 $\pm$ 1.0                   | 1.6 $\pm$ 0.2                   | 2.4 $\pm$ 0.2                    | 10.1 $\pm$ 0.5                   | 6.3 $\pm$ 0.3                    | 13.1 $\pm$ 0.6                   | 12.0 $\pm$ 0.7                   |
| DiffusionDB-10k | SimCLR              | 67.3 $\pm$ 0.1                   | 36.2 $\pm$ 0.4                   | 4.3 $\pm$ 0.2                   | 8.8 $\pm$ 0.1                    | 26.8 $\pm$ 0.4                   | 17.3 $\pm$ 0.1                   | 36.7 $\pm$ 0.8                   | 28.2 $\pm$ 0.3                   |
|                 | TS                  | 64.7 $\pm$ 0.4                   | 33.6 $\pm$ 0.3                   | 3.1 $\pm$ 0.5                   | 7.3 $\pm$ 0.3                    | 23.2 $\pm$ 0.4                   | 17.0 $\pm$ 0.0                   | 37.8 $\pm$ 0.6                   | 26.7 $\pm$ 0.4                   |
|                 | SDCLR               | 32.6 $\pm$ 1.2                   | 10.0 $\pm$ 0.3                   | 1.3 $\pm$ 0.1                   | 2.2 $\pm$ 0.2                    | 6.1 $\pm$ 0.7                    | 4.8 $\pm$ 0.4                    | 10.6 $\pm$ 0.5                   | 9.7 $\pm$ 0.5                    |
|                 | BRIDGE (ours)       | <b>68.3 <math>\pm</math> 0.2</b> | <b>37.4 <math>\pm</math> 0.4</b> | <b>5.4 <math>\pm</math> 0.3</b> | <b>9.8 <math>\pm</math> 0.1</b>  | <b>30.4 <math>\pm</math> 0.2</b> | <b>18.6 <math>\pm</math> 0.2</b> | <b>39.4 <math>\pm</math> 1.0</b> | <b>29.9 <math>\pm</math> 0.3</b> |
|                 | BRIDGE+TS (ours)    | 57.5 $\pm$ 0.8                   | 26.6 $\pm$ 0.3                   | 3.0 $\pm$ 0.4                   | 6.5 $\pm$ 0.1                    | 21.7 $\pm$ 0.8                   | 15.5 $\pm$ 0.4                   | 37.0 $\pm$ 0.4                   | 24.0 $\pm$ 0.5                   |
|                 | BRIDGE+SDCLR (ours) | 35.2 $\pm$ 0.7                   | 11.1 $\pm$ 0.3                   | 1.0 $\pm$ 0.1                   | 2.2 $\pm$ 0.1                    | 8.1 $\pm$ 1.0                    | 5.0 $\pm$ 0.3                    | 11.3 $\pm$ 0.7                   | 10.6 $\pm$ 0.5                   |

Table 10: Pretraining objective ablation on the default ResNet-50 backbone, full linear-probe results. DINO is excluded here because its reference recipe is defined for ViT backbones; the DINO comparison is reported with ViT-S in Table 3.

| Setting          | C10                              | C100                             | Cars                             | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| SimCLR (default) | <b>76.0 <math>\pm</math> 0.3</b> | <b>49.1 <math>\pm</math> 0.2</b> | <b>15.2 <math>\pm</math> 1.2</b> | <b>22.4 <math>\pm</math> 0.4</b> | <b>36.8 <math>\pm</math> 1.9</b> | <b>42.9 <math>\pm</math> 1.0</b> | <b>53.8 <math>\pm</math> 0.1</b> | <b>42.3 <math>\pm</math> 0.7</b> |
| MoCo             | 50.5 $\pm$ 1.0                   | 20.6 $\pm$ 2.7                   | 3.0 $\pm$ 0.2                    | 5.7 $\pm$ 0.5                    | 11.5 $\pm$ 3.0                   | 15.3 $\pm$ 0.3                   | 28.2 $\pm$ 0.7                   | 19.3 $\pm$ 1.2                   |

Table 11: Pretraining objective ablation on the default ResNet-50 backbone, full  $k$ NN results. DINO is excluded here because its reference recipe is defined for ViT backbones; the DINO comparison is reported with ViT-S in Table 3.

| Setting          | C10                              | C100                             | Cars                            | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|------------------|----------------------------------|----------------------------------|---------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| SimCLR (default) | <b>69.0 <math>\pm</math> 0.5</b> | <b>37.8 <math>\pm</math> 0.3</b> | <b>6.7 <math>\pm</math> 0.3</b> | <b>11.7 <math>\pm</math> 0.2</b> | <b>36.5 <math>\pm</math> 0.4</b> | <b>25.3 <math>\pm</math> 1.0</b> | <b>48.1 <math>\pm</math> 0.6</b> | <b>33.6 <math>\pm</math> 0.5</b> |
| MoCo             | 58.0 $\pm$ 0.4                   | 26.7 $\pm$ 0.5                   | 3.0 $\pm$ 0.2                   | 5.4 $\pm$ 0.2                    | 16.3 $\pm$ 0.9                   | 12.2 $\pm$ 0.6                   | 25.1 $\pm$ 0.5                   | 21.0 $\pm$ 0.5                   |

Table 12: Augmentation mechanism ablation with full linear-probe results. Oracle re-population is not a competing generator: it re-adds the exact real images withheld when the long tail was built, so it has access to the identity and location of the missing support, which the method only estimates. It is the upper bound on what any repair operator can achieve at this budget, and a generator that approaches it is recovering most of the benefit of the withheld real data without access to it.

| Setting                            | C10                              | C100                             | Cars                             | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Stable Diffusion 3 (default)       | <b>75.0 <math>\pm</math> 0.3</b> | <b>48.5 <math>\pm</math> 0.2</b> | <b>16.3 <math>\pm</math> 0.8</b> | <b>20.7 <math>\pm</math> 0.4</b> | 35.7 $\pm$ 0.9                   | <b>41.8 <math>\pm</math> 0.8</b> | <b>52.4 <math>\pm</math> 0.7</b> | <b>41.5 <math>\pm</math> 0.6</b> |
| FLUX                               | 70.7 $\pm$ 0.4                   | 42.1 $\pm$ 1.4                   | 10.3 $\pm$ 1.3                   | 17.5 $\pm$ 0.9                   | 29.4 $\pm$ 4.6                   | 31.5 $\pm$ 2.9                   | 43.7 $\pm$ 1.4                   | 35.0 $\pm$ 1.9                   |
| Oracle re-population (upper bound) | 74.8 $\pm$ 0.4                   | 47.8 $\pm$ 0.7                   | 13.1 $\pm$ 0.1                   | 20.5 $\pm$ 0.5                   | <b>40.3 <math>\pm</math> 6.1</b> | 38.1 $\pm$ 1.5                   | 50.3 $\pm$ 0.4                   | 40.7 $\pm$ 1.4                   |

Table 13: Augmentation mechanism ablation with full  $k$ NN results. Oracle re-population is not a competing generator: it re-adds the exact real images withheld when the long tail was built, so it has access to the identity and location of the missing support, which the method only estimates. It is the upper bound on what any repair operator can achieve at this budget, and a generator that approaches it is recovering most of the benefit of the withheld real data without access to it.

| Setting                            | C10                              | C100                             | Cars                            | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|------------------------------------|----------------------------------|----------------------------------|---------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Stable Diffusion 3 (default)       | 68.6 $\pm$ 0.1                   | 37.2 $\pm$ 0.2                   | 6.4 $\pm$ 0.6                   | <b>11.5 <math>\pm</math> 0.2</b> | <b>35.2 <math>\pm</math> 0.3</b> | <b>24.3 <math>\pm</math> 0.6</b> | <b>46.8 <math>\pm</math> 0.7</b> | <b>32.9 <math>\pm</math> 0.4</b> |
| FLUX                               | 65.8 $\pm$ 1.0                   | 35.4 $\pm$ 0.8                   | 5.7 $\pm$ 0.7                   | 9.4 $\pm$ 0.9                    | 27.7 $\pm$ 1.0                   | 18.8 $\pm$ 1.3                   | 38.7 $\pm$ 1.2                   | 28.8 $\pm$ 1.0                   |
| Oracle re-population (upper bound) | <b>68.8 <math>\pm</math> 0.2</b> | <b>37.7 <math>\pm</math> 0.3</b> | <b>6.6 <math>\pm</math> 0.5</b> | 11.4 $\pm$ 0.5                   | 32.7 $\pm$ 0.1                   | 23.6 $\pm$ 0.6                   | 44.4 $\pm$ 0.4                   | 32.2 $\pm$ 0.4                   |

Table 14: Selection rule ablation with full linear-probe results.

| Setting                                  | C10                              | C100                             | Cars                             | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|------------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Mode-window q1-99 + diversity            | 75.6 $\pm$ 0.1                   | 49.2 $\pm$ 0.2                   | <b>16.9 <math>\pm</math> 0.7</b> | 21.9 $\pm$ 0.4                   | 34.6 $\pm$ 2.9                   | 43.5 $\pm$ 0.7                   | 52.8 $\pm$ 0.1                   | 42.1 $\pm$ 0.7                   |
| Mode-window q50-99 + diversity           | 75.5 $\pm$ 0.1                   | <b>49.3 <math>\pm</math> 0.4</b> | 15.2 $\pm$ 0.3                   | <b>22.4 <math>\pm</math> 0.6</b> | 35.5 $\pm$ 1.1                   | <b>43.8 <math>\pm</math> 3.1</b> | 53.4 $\pm$ 0.9                   | 42.2 $\pm$ 0.9                   |
| Mode-window q75-99 + diversity (default) | <b>76.0 <math>\pm</math> 0.3</b> | 49.1 $\pm$ 0.2                   | 15.2 $\pm$ 1.2                   | 22.4 $\pm$ 0.4                   | <b>36.8 <math>\pm</math> 1.9</b> | 42.9 $\pm$ 1.0                   | <b>53.8 <math>\pm</math> 0.1</b> | <b>42.3 <math>\pm</math> 0.7</b> |
| Top-tail                                 | 70.8 $\pm$ 0.5                   | 43.1 $\pm$ 1.0                   | 10.0 $\pm$ 1.7                   | 15.9 $\pm$ 1.3                   | 27.7 $\pm$ 1.0                   | 30.5 $\pm$ 1.2                   | 42.2 $\pm$ 1.4                   | 34.3 $\pm$ 1.2                   |
| Uniform                                  | 71.5 $\pm$ 0.2                   | 43.8 $\pm$ 0.7                   | 9.3 $\pm$ 0.9                    | 17.5 $\pm$ 1.2                   | 32.9 $\pm$ 0.4                   | 31.7 $\pm$ 3.9                   | 42.0 $\pm$ 0.9                   | 35.5 $\pm$ 1.2                   |

Table 15: Selection rule ablation with full  $k$ NN results.

| Setting                                  | C10                              | C100                             | Cars                            | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|------------------------------------------|----------------------------------|----------------------------------|---------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Mode-window q1-99 + diversity            | <b>69.2 <math>\pm</math> 0.3</b> | 37.7 $\pm$ 0.2                   | 6.6 $\pm$ 0.2                   | 11.4 $\pm$ 0.2                   | 35.9 $\pm$ 0.2                   | 25.0 $\pm$ 0.5                   | 48.3 $\pm$ 0.3                   | 33.4 $\pm$ 0.3                   |
| Mode-window q50-99 + diversity           | 68.7 $\pm$ 0.1                   | <b>38.0 <math>\pm</math> 0.2</b> | <b>7.3 <math>\pm</math> 0.2</b> | 11.3 $\pm$ 0.2                   | <b>36.5 <math>\pm</math> 0.1</b> | 24.8 $\pm$ 0.8                   | <b>48.5 <math>\pm</math> 0.4</b> | 33.6 $\pm$ 0.3                   |
| Mode-window q75-99 + diversity (default) | 69.0 $\pm$ 0.5                   | 37.8 $\pm$ 0.3                   | 6.7 $\pm$ 0.3                   | <b>11.7 <math>\pm</math> 0.2</b> | 36.5 $\pm$ 0.4                   | <b>25.3 <math>\pm</math> 1.0</b> | 48.1 $\pm$ 0.6                   | <b>33.6 <math>\pm</math> 0.5</b> |
| Top-tail                                 | 65.7 $\pm$ 0.8                   | 34.4 $\pm$ 0.2                   | 4.1 $\pm$ 0.4                   | 9.6 $\pm$ 0.2                    | 23.9 $\pm$ 0.9                   | 18.1 $\pm$ 0.8                   | 35.4 $\pm$ 1.2                   | 27.3 $\pm$ 0.6                   |
| Uniform                                  | 66.1 $\pm$ 1.0                   | 35.7 $\pm$ 0.3                   | 4.9 $\pm$ 0.1                   | 10.2 $\pm$ 0.7                   | 25.3 $\pm$ 0.4                   | 19.0 $\pm$ 0.8                   | 37.4 $\pm$ 0.6                   | 28.4 $\pm$ 0.6                   |

Table 16: Cycle count ablation with full linear-probe results.

| Setting            | C10                              | C100                             | Cars                             | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|--------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| 2 cycles           | 68.9 $\pm$ 0.6                   | 41.6 $\pm$ 1.3                   | 10.6 $\pm$ 1.2                   | 16.4 $\pm$ 0.4                   | 30.7 $\pm$ 4.5                   | 33.2 $\pm$ 0.9                   | 43.3 $\pm$ 0.6                   | 35.0 $\pm$ 1.4                   |
| 5 cycles (default) | <b>76.0 <math>\pm</math> 0.3</b> | <b>49.1 <math>\pm</math> 0.2</b> | <b>15.2 <math>\pm</math> 1.2</b> | <b>22.4 <math>\pm</math> 0.4</b> | <b>36.8 <math>\pm</math> 1.9</b> | <b>42.9 <math>\pm</math> 1.0</b> | <b>53.8 <math>\pm</math> 0.1</b> | <b>42.3 <math>\pm</math> 0.7</b> |
| 10 cycles          | 70.5 $\pm$ 1.2                   | 43.8 $\pm$ 0.6                   | 9.2 $\pm$ 0.9                    | 17.2 $\pm$ 1.6                   | 24.8 $\pm$ 5.2                   | 31.3 $\pm$ 1.7                   | 41.9 $\pm$ 0.6                   | 34.1 $\pm$ 1.7                   |

Table 17: Cycle count ablation with full  $k$ NN results.

| Setting            | C10                              | C100                             | Cars                            | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|--------------------|----------------------------------|----------------------------------|---------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| 2 cycles           | 65.3 $\pm$ 0.4                   | 34.2 $\pm$ 0.5                   | 5.0 $\pm$ 0.7                   | 10.0 $\pm$ 0.2                   | 26.1 $\pm$ 0.8                   | 18.7 $\pm$ 0.6                   | 35.9 $\pm$ 0.3                   | 27.9 $\pm$ 0.5                   |
| 5 cycles (default) | <b>69.0 <math>\pm</math> 0.5</b> | <b>37.8 <math>\pm</math> 0.3</b> | <b>6.7 <math>\pm</math> 0.3</b> | <b>11.7 <math>\pm</math> 0.2</b> | <b>36.5 <math>\pm</math> 0.4</b> | <b>25.3 <math>\pm</math> 1.0</b> | <b>48.1 <math>\pm</math> 0.6</b> | <b>33.6 <math>\pm</math> 0.5</b> |
| 10 cycles          | 66.4 $\pm$ 0.5                   | 35.4 $\pm$ 0.3                   | 4.8 $\pm$ 0.3                   | 10.4 $\pm$ 0.6                   | 25.2 $\pm$ 0.8                   | 18.6 $\pm$ 0.9                   | 36.0 $\pm$ 0.7                   | 28.1 $\pm$ 0.6                   |

Table 18: Backbone architecture ablation with full linear-probe results.

| Setting             | C10                              | C100                             | Cars                             | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|---------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| ResNet-18           | 68.8 $\pm$ 0.1                   | 41.7 $\pm$ 0.2                   | 13.5 $\pm$ 0.3                   | 19.7 $\pm$ 0.6                   | 38.7 $\pm$ 3.4                   | 38.4 $\pm$ 2.6                   | 50.2 $\pm$ 0.2                   | 38.7 $\pm$ 1.1                   |
| ResNet-50 (default) | <b>76.0 <math>\pm</math> 0.3</b> | <b>49.1 <math>\pm</math> 0.2</b> | <b>15.2 <math>\pm</math> 1.2</b> | <b>22.4 <math>\pm</math> 0.4</b> | 36.8 $\pm$ 1.9                   | <b>42.9 <math>\pm</math> 1.0</b> | <b>53.8 <math>\pm</math> 0.1</b> | <b>42.3 <math>\pm</math> 0.7</b> |
| ViT-S               | 62.7 $\pm$ 0.4                   | 37.6 $\pm$ 0.6                   | 8.8 $\pm$ 0.5                    | 12.2 $\pm$ 0.2                   | 35.1 $\pm$ 4.2                   | 30.2 $\pm$ 1.6                   | 42.5 $\pm$ 1.2                   | 32.7 $\pm$ 1.3                   |
| ViT-B               | 66.0 $\pm$ 2.3                   | 40.6 $\pm$ 2.3                   | 8.3 $\pm$ 0.5                    | 14.3 $\pm$ 0.8                   | <b>38.8 <math>\pm</math> 2.6</b> | 31.3 $\pm$ 2.1                   | 44.5 $\pm$ 1.7                   | 34.8 $\pm$ 1.8                   |

Table 20: Selector robustness on ImageNet-100-LT. Each row changes one component of the acquisition rule – candidate pool multiplier, upper cutoff, neighbourhood size, distance, or selection strategy – and holds the rest of the loop fixed. Mean  $\pm$  standard deviation over three seeds. Cosine scoring rescales the normalized  $L_2$  score by a positive constant and therefore selects the identical anchor set; that row is a null control, and its distance from the normalized  $L_2$  row measures the end-to-end run-to-run spread of the pipeline. Linear-probe accuracy.

| Method                                | Cars            | Aircraft         | Flowers          | Pets             | CIFAR-10         | CIFAR-100        | IN100-LT         |
|---------------------------------------|-----------------|------------------|------------------|------------------|------------------|------------------|------------------|
| Candidate pool $\alpha = 1$           | 6.46 $\pm$ 0.57 | 11.37 $\pm$ 1.59 | 23.53 $\pm$ 3.00 | 21.92 $\pm$ 2.00 | 61.94 $\pm$ 0.94 | 32.73 $\pm$ 0.21 | 33.92 $\pm$ 1.68 |
| Candidate pool $\alpha = 2$           | 6.77 $\pm$ 0.63 | 11.59 $\pm$ 0.40 | 21.79 $\pm$ 3.22 | 23.19 $\pm$ 3.93 | 62.90 $\pm$ 1.06 | 34.48 $\pm$ 0.11 | 35.69 $\pm$ 0.87 |
| Candidate pool $\alpha = 4$ (default) | 6.31 $\pm$ 0.10 | 11.17 $\pm$ 3.78 | 23.53 $\pm$ 3.27 | 24.18 $\pm$ 2.16 | 62.11 $\pm$ 0.19 | 33.39 $\pm$ 0.64 | 34.18 $\pm$ 0.47 |
| Candidate pool $\alpha = 8$           | 6.41 $\pm$ 0.28 | 11.16 $\pm$ 0.78 | 23.09 $\pm$ 4.35 | 24.82 $\pm$ 3.29 | 62.60 $\pm$ 1.79 | 33.15 $\pm$ 1.01 | 35.40 $\pm$ 0.87 |
| Upper cutoff $q = 0.95$               | 6.79 $\pm$ 0.65 | 11.44 $\pm$ 0.85 | 19.83 $\pm$ 2.47 | 25.27 $\pm$ 2.32 | 62.57 $\pm$ 0.87 | 33.83 $\pm$ 0.53 | 35.13 $\pm$ 1.03 |
| Upper cutoff $q = 0.99$ (default)     | 6.59 $\pm$ 0.17 | 11.76 $\pm$ 1.54 | 20.26 $\pm$ 1.73 | 24.00 $\pm$ 2.20 | 63.11 $\pm$ 0.75 | 33.43 $\pm$ 1.02 | 35.85 $\pm$ 1.15 |
| Upper cutoff $q = 0.995$              | 6.78 $\pm$ 0.66 | 10.40 $\pm$ 2.49 | 22.88 $\pm$ 4.58 | 22.55 $\pm$ 3.09 | 62.76 $\pm$ 0.50 | 33.87 $\pm$ 0.38 | 35.32 $\pm$ 0.64 |
| Upper cutoff $q = 1.0$                | 6.31 $\pm$ 0.45 | 10.94 $\pm$ 1.34 | 22.44 $\pm$ 1.89 | 23.37 $\pm$ 4.00 | 62.85 $\pm$ 1.04 | 33.91 $\pm$ 0.87 | 35.51 $\pm$ 1.59 |
| Neighbourhood $k = 10$                | 6.26 $\pm$ 0.29 | 10.44 $\pm$ 1.43 | 22.22 $\pm$ 0.65 | 24.00 $\pm$ 1.23 | 62.66 $\pm$ 0.60 | 33.44 $\pm$ 0.58 | 35.14 $\pm$ 1.21 |
| Neighbourhood $k = 25$                | 6.44 $\pm$ 0.76 | 11.13 $\pm$ 1.19 | 19.39 $\pm$ 3.60 | 23.37 $\pm$ 1.18 | 63.18 $\pm$ 0.56 | 33.40 $\pm$ 1.21 | 33.49 $\pm$ 0.43 |
| Neighbourhood $k = 50$                | 6.34 $\pm$ 0.56 | 10.01 $\pm$ 1.32 | 20.48 $\pm$ 1.00 | 22.64 $\pm$ 3.24 | 61.28 $\pm$ 2.19 | 32.64 $\pm$ 0.29 | 34.23 $\pm$ 0.77 |
| Neighbourhood $k = 200$               | 6.50 $\pm$ 0.46 | 10.58 $\pm$ 2.83 | 20.26 $\pm$ 1.13 | 22.46 $\pm$ 1.23 | 61.24 $\pm$ 1.08 | 32.87 $\pm$ 0.76 | 34.42 $\pm$ 1.23 |
| Distance: normalized $L_2$ (default)  | 6.59 $\pm$ 0.17 | 11.79 $\pm$ 1.34 | 20.48 $\pm$ 2.64 | 24.46 $\pm$ 2.68 | 61.69 $\pm$ 0.67 | 33.88 $\pm$ 1.05 | 35.24 $\pm$ 1.10 |
| Distance: cosine (null control)       | 6.55 $\pm$ 0.15 | 10.50 $\pm$ 1.10 | 20.04 $\pm$ 2.64 | 23.91 $\pm$ 1.70 | 62.22 $\pm$ 1.29 | 33.23 $\pm$ 0.81 | 34.98 $\pm$ 0.51 |
| Distance: raw $L_2$                   | 6.92 $\pm$ 0.78 | 10.45 $\pm$ 3.00 | 18.95 $\pm$ 5.23 | 25.72 $\pm$ 2.31 | 62.81 $\pm$ 0.80 | 33.20 $\pm$ 0.89 | 35.08 $\pm$ 2.48 |
| Strategy: mode window (default)       | 6.55 $\pm$ 0.15 | 10.18 $\pm$ 1.30 | 22.44 $\pm$ 4.59 | 23.28 $\pm$ 2.04 | 60.95 $\pm$ 3.01 | 33.05 $\pm$ 0.21 | 34.53 $\pm$ 0.92 |
| Strategy: sparse band with FPS        | 6.60 $\pm$ 0.62 | 11.21 $\pm$ 1.26 | 21.13 $\pm$ 2.64 | 22.46 $\pm$ 0.16 | 63.34 $\pm$ 0.27 | 34.29 $\pm$ 0.30 | 34.30 $\pm$ 1.26 |
| Strategy: sparse band, random         | 6.46 $\pm$ 0.48 | 11.06 $\pm$ 0.70 | 21.35 $\pm$ 4.82 | 22.92 $\pm$ 0.87 | 62.26 $\pm$ 0.89 | 32.33 $\pm$ 0.96 | 34.46 $\pm$ 1.08 |
| Strategy: inverse-cluster             | 6.44 $\pm$ 0.75 | 10.66 $\pm$ 1.06 | 19.17 $\pm$ 2.47 | 21.65 $\pm$ 2.28 | 62.70 $\pm$ 0.45 | 33.50 $\pm$ 0.95 | 34.69 $\pm$ 1.10 |
| Strategy: densest window              | 6.64 $\pm$ 0.46 | 11.90 $\pm$ 0.49 | 20.04 $\pm$ 1.00 | 24.28 $\pm$ 1.64 | 63.50 $\pm$ 0.60 | 33.80 $\pm$ 0.93 | 35.20 $\pm$ 0.80 |
| Strategy: top tail, no diversity      | 6.39 $\pm$ 0.43 | 10.48 $\pm$ 1.03 | 19.83 $\pm$ 2.00 | 23.82 $\pm$ 1.50 | 62.86 $\pm$ 0.92 | 33.14 $\pm$ 1.02 | 34.66 $\pm$ 0.41 |

Table 21: Selector robustness on ImageNet-100-LT. Each row changes one component of the acquisition rule – candidate pool multiplier, upper cutoff, neighbourhood size, distance, or selection strategy – and holds the rest of the loop fixed. Mean  $\pm$  standard deviation over three seeds. Cosine scoring rescales the normalized  $L_2$  score by a positive constant and therefore selects the identical anchor set; that row is a null control, and its distance from the normalized  $L_2$  row measures the end-to-end run-to-run spread of the pipeline.  $k$ NN accuracy.

| Method                                | Cars            | Aircraft        | Flowers          | Pets             | CIFAR-10         | CIFAR-100        | IN100-LT         |
|---------------------------------------|-----------------|-----------------|------------------|------------------|------------------|------------------|------------------|
| Candidate pool $\alpha = 1$           | 5.02 $\pm$ 0.14 | 8.62 $\pm$ 0.26 | 19.22 $\pm$ 0.71 | 14.92 $\pm$ 0.99 | 62.40 $\pm$ 0.12 | 31.47 $\pm$ 0.25 | 33.47 $\pm$ 1.05 |
| Candidate pool $\alpha = 2$           | 5.08 $\pm$ 0.15 | 8.36 $\pm$ 0.22 | 19.07 $\pm$ 0.42 | 15.17 $\pm$ 0.27 | 62.81 $\pm$ 0.58 | 31.72 $\pm$ 0.27 | 33.19 $\pm$ 0.61 |
| Candidate pool $\alpha = 4$ (default) | 4.92 $\pm$ 0.15 | 8.48 $\pm$ 0.72 | 19.04 $\pm$ 0.03 | 14.60 $\pm$ 0.38 | 62.35 $\pm$ 0.54 | 31.84 $\pm$ 0.37 | 32.59 $\pm$ 0.49 |
| Candidate pool $\alpha = 8$           | 4.85 $\pm$ 0.17 | 8.53 $\pm$ 0.41 | 19.17 $\pm$ 0.90 | 14.82 $\pm$ 0.62 | 62.15 $\pm$ 0.07 | 31.56 $\pm$ 0.23 | 33.58 $\pm$ 1.03 |
| Upper cutoff $q = 0.95$               | 4.99 $\pm$ 0.11 | 8.57 $\pm$ 0.51 | 19.09 $\pm$ 0.51 | 14.68 $\pm$ 0.28 | 62.84 $\pm$ 0.45 | 31.88 $\pm$ 0.04 | 32.56 $\pm$ 0.45 |
| Upper cutoff $q = 0.99$ (default)     | 4.95 $\pm$ 0.17 | 8.53 $\pm$ 0.69 | 19.40 $\pm$ 0.45 | 14.85 $\pm$ 0.49 | 62.29 $\pm$ 0.50 | 31.87 $\pm$ 0.16 | 32.44 $\pm$ 0.48 |
| Upper cutoff $q = 0.995$              | 5.01 $\pm$ 0.07 | 8.34 $\pm$ 0.74 | 19.58 $\pm$ 0.83 | 14.87 $\pm$ 0.48 | 63.05 $\pm$ 0.47 | 32.06 $\pm$ 0.11 | 33.01 $\pm$ 0.34 |
| Upper cutoff $q = 1.0$                | 4.99 $\pm$ 0.08 | 8.15 $\pm$ 0.61 | 18.52 $\pm$ 0.98 | 14.45 $\pm$ 0.19 | 62.84 $\pm$ 0.32 | 31.69 $\pm$ 0.30 | 32.99 $\pm$ 0.83 |
| Neighbourhood $k = 10$                | 5.04 $\pm$ 0.09 | 8.99 $\pm$ 1.21 | 18.86 $\pm$ 0.35 | 14.90 $\pm$ 0.57 | 62.73 $\pm$ 0.39 | 31.69 $\pm$ 0.17 | 33.14 $\pm$ 0.40 |
| Neighbourhood $k = 25$                | 4.97 $\pm$ 0.14 | 8.59 $\pm$ 0.21 | 19.05 $\pm$ 0.29 | 15.04 $\pm$ 0.85 | 62.68 $\pm$ 0.66 | 31.82 $\pm$ 0.16 | 33.56 $\pm$ 0.97 |
| Neighbourhood $k = 50$                | 5.06 $\pm$ 0.17 | 8.62 $\pm$ 0.77 | 19.26 $\pm$ 0.67 | 14.92 $\pm$ 0.03 | 62.63 $\pm$ 0.40 | 31.71 $\pm$ 0.59 | 31.88 $\pm$ 0.57 |
| Neighbourhood $k = 200$               | 5.17 $\pm$ 0.15 | 8.65 $\pm$ 0.59 | 19.22 $\pm$ 1.02 | 15.17 $\pm$ 0.10 | 62.80 $\pm$ 0.31 | 31.90 $\pm$ 0.13 | 33.91 $\pm$ 0.54 |
| Distance: normalized $L_2$ (default)  | 4.97 $\pm$ 0.18 | 8.55 $\pm$ 0.67 | 19.42 $\pm$ 0.49 | 14.85 $\pm$ 0.50 | 62.27 $\pm$ 0.52 | 31.86 $\pm$ 0.16 | 32.60 $\pm$ 0.96 |
| Distance: cosine (null control)       | 4.90 $\pm$ 0.13 | 8.52 $\pm$ 0.69 | 19.35 $\pm$ 0.54 | 14.75 $\pm$ 0.44 | 62.31 $\pm$ 0.53 | 31.78 $\pm$ 0.28 | 32.44 $\pm$ 0.83 |
| Distance: raw $L_2$                   | 4.85 $\pm$ 0.10 | 7.83 $\pm$ 0.39 | 19.13 $\pm$ 0.88 | 14.63 $\pm$ 0.34 | 62.81 $\pm$ 0.52 | 31.71 $\pm$ 0.04 | 32.28 $\pm$ 0.65 |
| Strategy: mode window (default)       | 4.90 $\pm$ 0.12 | 8.52 $\pm$ 0.69 | 19.34 $\pm$ 0.54 | 14.75 $\pm$ 0.45 | 62.31 $\pm$ 0.52 | 31.78 $\pm$ 0.28 | 32.73 $\pm$ 0.75 |
| Strategy: sparse band with FPS        | 5.19 $\pm$ 0.04 | 8.12 $\pm$ 0.39 | 19.53 $\pm$ 0.49 | 14.95 $\pm$ 0.75 | 62.43 $\pm$ 0.17 | 31.62 $\pm$ 0.32 | 33.38 $\pm$ 1.46 |
| Strategy: sparse band, random         | 5.00 $\pm$ 0.13 | 8.70 $\pm$ 0.06 | 19.36 $\pm$ 0.88 | 14.95 $\pm$ 0.61 | 62.63 $\pm$ 0.19 | 31.64 $\pm$ 0.62 | 33.04 $\pm$ 0.82 |
| Strategy: inverse-cluster             | 5.05 $\pm$ 0.02 | 8.83 $\pm$ 1.05 | 19.40 $\pm$ 0.81 | 15.06 $\pm$ 0.46 | 62.60 $\pm$ 0.50 | 31.99 $\pm$ 0.37 | 33.75 $\pm$ 0.44 |
| Strategy: densest window              | 4.88 $\pm$ 0.10 | 8.44 $\pm$ 0.54 | 19.34 $\pm$ 0.59 | 14.95 $\pm$ 0.53 | 62.46 $\pm$ 0.12 | 31.62 $\pm$ 0.30 | 33.08 $\pm$ 0.64 |
| Strategy: top tail, no diversity      | 4.90 $\pm$ 0.12 | 8.38 $\pm$ 0.29 | 19.11 $\pm$ 0.64 | 15.22 $\pm$ 0.03 | 62.05 $\pm$ 0.77 | 31.77 $\pm$ 0.13 | 32.41 $\pm$ 0.25 |

Table 19: Backbone architecture ablation with full  $k$ NN results.

| Setting             | C10                              | C100                             | Cars                            | Aircraft                         | Flowers                          | Pets                             | IN100-LT                         | Avg.                             |
|---------------------|----------------------------------|----------------------------------|---------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| ResNet-18           | 66.3 $\pm$ 0.2                   | 35.7 $\pm$ 0.3                   | 5.9 $\pm$ 0.2                   | 11.1 $\pm$ 0.4                   | 33.5 $\pm$ 0.1                   | 23.3 $\pm$ 0.5                   | 44.5 $\pm$ 0.7                   | 31.5 $\pm$ 0.4                   |
| ResNet-50 (default) | 69.0 $\pm$ 0.5                   | 37.8 $\pm$ 0.3                   | <b>6.7 <math>\pm</math> 0.3</b> | <b>11.7 <math>\pm</math> 0.2</b> | 36.5 $\pm$ 0.4                   | <b>25.3 <math>\pm</math> 1.0</b> | <b>48.1 <math>\pm</math> 0.6</b> | <b>33.6 <math>\pm</math> 0.5</b> |
| ViT-S               | 67.3 $\pm$ 0.4                   | 37.5 $\pm$ 0.3                   | 5.3 $\pm$ 0.2                   | 7.6 $\pm$ 0.1                    | 36.2 $\pm$ 0.7                   | 20.1 $\pm$ 1.0                   | 42.1 $\pm$ 1.0                   | 30.9 $\pm$ 0.5                   |
| ViT-B               | <b>69.3 <math>\pm</math> 0.8</b> | <b>40.1 <math>\pm</math> 0.8</b> | 6.0 $\pm$ 0.2                   | 8.3 $\pm$ 0.2                    | <b>39.8 <math>\pm</math> 0.8</b> | 21.9 $\pm$ 1.4                   | 45.1 $\pm$ 0.7                   | 32.9 $\pm$ 0.7                   |

Table 22: Repair utility by sparsity band on ImageNet-100-LT. Each row spends the identical acquisition budget on a different quantile range of the  $k$ NN score distribution, holding every other part of the loop fixed. Mean  $\pm$  standard deviation over three seeds. Linear-probe accuracy.

| Method                 | Cars            | Aircraft         | Flowers          | Pets             | CIFAR-10         | CIFAR-100        | IN100-LT         |
|------------------------|-----------------|------------------|------------------|------------------|------------------|------------------|------------------|
| q0–25 (densest)        | 6.73 $\pm$ 0.36 | 9.63 $\pm$ 0.26  | 24.84 $\pm$ 5.10 | 23.37 $\pm$ 2.72 | 62.97 $\pm$ 0.82 | 33.96 $\pm$ 0.49 | 34.44 $\pm$ 0.62 |
| q25–50                 | 6.80 $\pm$ 0.39 | 10.28 $\pm$ 1.37 | 23.75 $\pm$ 2.10 | 22.74 $\pm$ 1.57 | 62.76 $\pm$ 0.64 | 33.20 $\pm$ 1.12 | 35.61 $\pm$ 0.23 |
| q50–75                 | 6.79 $\pm$ 0.63 | 10.24 $\pm$ 1.94 | 20.04 $\pm$ 4.64 | 23.55 $\pm$ 2.58 | 62.24 $\pm$ 0.37 | 32.80 $\pm$ 1.66 | 34.51 $\pm$ 1.27 |
| q75–85                 | 6.12 $\pm$ 0.33 | 9.74 $\pm$ 1.00  | 22.88 $\pm$ 4.58 | 23.46 $\pm$ 1.64 | 60.97 $\pm$ 1.05 | 31.90 $\pm$ 0.36 | 33.55 $\pm$ 0.46 |
| q85–90                 | 6.62 $\pm$ 0.39 | 9.95 $\pm$ 0.73  | 23.75 $\pm$ 3.09 | 25.45 $\pm$ 1.77 | 63.38 $\pm$ 0.78 | 33.47 $\pm$ 0.93 | 35.24 $\pm$ 0.69 |
| q90–95                 | 6.55 $\pm$ 0.55 | 10.46 $\pm$ 1.87 | 22.66 $\pm$ 3.94 | 25.27 $\pm$ 0.72 | 62.30 $\pm$ 1.15 | 33.49 $\pm$ 0.87 | 34.40 $\pm$ 0.62 |
| q95–97                 | 6.58 $\pm$ 0.20 | 9.55 $\pm$ 1.62  | 23.09 $\pm$ 1.00 | 23.55 $\pm$ 0.68 | 61.77 $\pm$ 2.13 | 32.75 $\pm$ 2.27 | 34.99 $\pm$ 1.19 |
| q97–99                 | 6.80 $\pm$ 0.17 | 11.38 $\pm$ 2.82 | 20.26 $\pm$ 3.98 | 26.27 $\pm$ 3.24 | 62.73 $\pm$ 1.49 | 33.13 $\pm$ 1.97 | 35.31 $\pm$ 1.55 |
| q99–100 (extreme tail) | 6.55 $\pm$ 0.28 | 10.78 $\pm$ 1.23 | 21.79 $\pm$ 0.75 | 23.82 $\pm$ 1.37 | 63.54 $\pm$ 0.92 | 33.65 $\pm$ 0.55 | 35.57 $\pm$ 0.91 |

Table 23: Repair utility by sparsity band on ImageNet-100-LT. Each row spends the identical acquisition budget on a different quantile range of the  $k$ NN score distribution, holding every other part of the loop fixed. Mean  $\pm$  standard deviation over three seeds.  $k$ NN accuracy.

| Method                 | Cars            | Aircraft        | Flowers          | Pets             | CIFAR-10         | CIFAR-100        | IN100-LT         |
|------------------------|-----------------|-----------------|------------------|------------------|------------------|------------------|------------------|
| q0–25 (densest)        | 4.97 $\pm$ 0.07 | 8.33 $\pm$ 0.72 | 20.00 $\pm$ 0.65 | 14.48 $\pm$ 0.27 | 62.04 $\pm$ 0.84 | 31.45 $\pm$ 0.05 | 31.88 $\pm$ 0.91 |
| q25–50                 | 5.16 $\pm$ 0.14 | 8.52 $\pm$ 0.37 | 19.58 $\pm$ 0.49 | 14.82 $\pm$ 0.73 | 62.95 $\pm$ 0.79 | 31.54 $\pm$ 0.09 | 32.88 $\pm$ 0.78 |
| q50–75                 | 5.06 $\pm$ 0.08 | 8.51 $\pm$ 0.44 | 19.63 $\pm$ 0.75 | 14.46 $\pm$ 0.81 | 62.41 $\pm$ 0.61 | 31.47 $\pm$ 0.56 | 32.61 $\pm$ 0.55 |
| q75–85                 | 4.97 $\pm$ 0.08 | 8.78 $\pm$ 0.45 | 19.54 $\pm$ 0.14 | 15.39 $\pm$ 0.20 | 62.52 $\pm$ 0.56 | 31.31 $\pm$ 0.43 | 32.23 $\pm$ 1.17 |
| q85–90                 | 5.04 $\pm$ 0.09 | 9.04 $\pm$ 0.48 | 19.61 $\pm$ 0.46 | 14.75 $\pm$ 0.65 | 62.57 $\pm$ 0.27 | 31.74 $\pm$ 0.34 | 32.82 $\pm$ 0.96 |
| q90–95                 | 4.92 $\pm$ 0.21 | 8.67 $\pm$ 0.75 | 19.60 $\pm$ 0.53 | 14.96 $\pm$ 0.46 | 62.40 $\pm$ 0.59 | 31.56 $\pm$ 0.54 | 32.48 $\pm$ 0.56 |
| q95–97                 | 4.88 $\pm$ 0.35 | 8.51 $\pm$ 1.09 | 19.40 $\pm$ 0.72 | 14.97 $\pm$ 0.49 | 62.32 $\pm$ 0.94 | 31.55 $\pm$ 0.43 | 31.63 $\pm$ 0.32 |
| q97–99                 | 5.02 $\pm$ 0.08 | 8.59 $\pm$ 0.43 | 19.63 $\pm$ 0.67 | 14.89 $\pm$ 0.56 | 62.64 $\pm$ 0.56 | 31.75 $\pm$ 0.61 | 32.59 $\pm$ 0.96 |
| q99–100 (extreme tail) | 5.17 $\pm$ 0.16 | 8.16 $\pm$ 0.23 | 19.10 $\pm$ 0.60 | 15.00 $\pm$ 0.53 | 62.81 $\pm$ 0.26 | 32.12 $\pm$ 0.43 | 32.15 $\pm$ 1.40 |

## I ALGORITHM

### Algorithm 1 BRIDGE training loop with mode-window selection

```

1: Initialize source set  $D^{(0)}$ , encoder  $f_{\theta^{(0)}}$ , frozen generator  $g_\phi$ , cycle budgets  $\{E_c\}_{c=0}^{C-1}$ 
2: for  $c = 0, \dots, C - 1$  do
3:   Train or continue training  $f_{\theta^{(c)}}$  on  $D^{(c)}$  for  $E_c$  epochs
4:   Compute embeddings  $Z^{(c)} = \{f_{\theta^{(c)}}(x_i) : x_i \in D^{(c)}\}$ 
5:   Compute mean  $k$ NN scores  $\{d_i^{(c)}\}_{i=1}^{N^{(c)}}$  from  $Z^{(c)}$ 
6:   Restrict  $\{d_i^{(c)}\}$  to the 75th to 99th percentile range and build a histogram
7:   Let  $m^{(c)}$  be the center of the modal histogram bin inside that upper-tail band
8:   Form a candidate set  $C^{(c)}$  of size  $4B$  whose scores are closest to  $m^{(c)}$ 
9:   Select  $S^{(c)} \subseteq C^{(c)}$  with  $|S^{(c)}| = B$  by greedy farthest-point sampling in embedding space
10:  if  $c < C - 1$  then
11:    Form  $\tilde{D}^{(c)} = \{g_\phi(x_i, \epsilon_j) \mid i \in S^{(c)}, j \in [G]\}$ 
12:    Update  $D^{(c+1)} = D^{(c)} \cup \tilde{D}^{(c)}$ 
13:  end if
14: end for
15: Return the final encoder and repaired source set

```

Table 24: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both arms of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Four objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO) and masked-reconstruction (MAE) pretraining. The first three run five cycles of 100 epochs and MAE five of 80. Mean  $\pm$  standard deviation over three seeds. Linear-probe accuracy.

| Method                | Cars            | Aircraft         | Flowers          | Pets             | CIFAR-10         | CIFAR-100        | IN100-LT         |
|-----------------------|-----------------|------------------|------------------|------------------|------------------|------------------|------------------|
| SimCLR, SSL baseline  | 7.15 $\pm$ 0.19 | 10.92 $\pm$ 0.40 | 32.90 $\pm$ 2.72 | 25.00 $\pm$ 1.25 | 60.46 $\pm$ 0.46 | 34.96 $\pm$ 0.20 | 41.95 $\pm$ 1.43 |
| SimCLR, with BRIDGE   | 8.18 $\pm$ 0.42 | 11.82 $\pm$ 0.43 | 36.60 $\pm$ 2.85 | 29.80 $\pm$ 2.98 | 60.85 $\pm$ 0.35 | 36.33 $\pm$ 0.14 | 43.93 $\pm$ 1.09 |
| MoCo v3, SSL baseline | 7.57 $\pm$ 0.23 | 11.05 $\pm$ 1.31 | 30.28 $\pm$ 1.36 | 29.26 $\pm$ 3.13 | 59.96 $\pm$ 0.39 | 34.47 $\pm$ 0.01 | 44.79 $\pm$ 1.17 |
| MoCo v3, with BRIDGE  | 9.55 $\pm$ 0.34 | 13.42 $\pm$ 0.98 | 36.82 $\pm$ 6.07 | 31.25 $\pm$ 4.00 | 62.36 $\pm$ 0.46 | 36.30 $\pm$ 0.13 | 47.55 $\pm$ 0.86 |
| DINO, SSL baseline    | 8.10 $\pm$ 0.25 | 11.40 $\pm$ 1.03 | 38.13 $\pm$ 5.24 | 27.54 $\pm$ 2.34 | 58.93 $\pm$ 0.94 | 33.56 $\pm$ 1.75 | 44.63 $\pm$ 1.47 |
| DINO, with BRIDGE     | 9.49 $\pm$ 0.88 | 13.28 $\pm$ 0.58 | 37.25 $\pm$ 2.61 | 34.60 $\pm$ 3.62 | 61.30 $\pm$ 0.07 | 35.23 $\pm$ 0.31 | 46.78 $\pm$ 0.77 |
| MAE, SSL baseline     | 1.54 $\pm$ 0.19 | 2.44 $\pm$ 1.00  | 10.46 $\pm$ 5.88 | 10.05 $\pm$ 3.03 | 37.69 $\pm$ 3.73 | 13.20 $\pm$ 1.94 | 19.93 $\pm$ 2.13 |
| MAE, with BRIDGE      | 1.56 $\pm$ 0.43 | 3.18 $\pm$ 0.70  | 8.50 $\pm$ 2.26  | 8.61 $\pm$ 2.20  | 41.46 $\pm$ 2.20 | 15.00 $\pm$ 2.40 | 22.19 $\pm$ 2.35 |

Table 25: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both arms of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Four objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO) and masked-reconstruction (MAE) pretraining. The first three run five cycles of 100 epochs and MAE five of 80. Mean  $\pm$  standard deviation over three seeds.  $k$ NN accuracy.

| Method                | Cars            | Aircraft        | Flowers          | Pets             | CIFAR-10         | CIFAR-100        | IN100-LT         |
|-----------------------|-----------------|-----------------|------------------|------------------|------------------|------------------|------------------|
| SimCLR, SSL baseline  | 6.26 $\pm$ 0.17 | 7.09 $\pm$ 0.17 | 36.53 $\pm$ 0.89 | 19.16 $\pm$ 0.64 | 66.05 $\pm$ 0.48 | 36.72 $\pm$ 0.36 | 41.21 $\pm$ 0.72 |
| SimCLR, with BRIDGE   | 6.52 $\pm$ 0.22 | 7.56 $\pm$ 0.23 | 38.29 $\pm$ 0.83 | 19.93 $\pm$ 0.53 | 65.70 $\pm$ 0.85 | 36.46 $\pm$ 0.33 | 42.91 $\pm$ 0.78 |
| MoCo v3, SSL baseline | 6.65 $\pm$ 0.38 | 7.71 $\pm$ 0.16 | 38.91 $\pm$ 1.03 | 23.56 $\pm$ 1.12 | 69.29 $\pm$ 0.25 | 39.98 $\pm$ 0.44 | 45.14 $\pm$ 0.46 |
| MoCo v3, with BRIDGE  | 6.74 $\pm$ 0.26 | 8.21 $\pm$ 0.18 | 40.51 $\pm$ 1.45 | 24.91 $\pm$ 0.45 | 65.41 $\pm$ 0.21 | 36.24 $\pm$ 0.46 | 47.71 $\pm$ 1.53 |
| DINO, SSL baseline    | 6.05 $\pm$ 0.27 | 7.73 $\pm$ 0.52 | 38.52 $\pm$ 1.19 | 19.26 $\pm$ 0.16 | 62.69 $\pm$ 2.42 | 33.98 $\pm$ 2.80 | 45.28 $\pm$ 0.98 |
| DINO, with BRIDGE     | 6.39 $\pm$ 0.22 | 8.16 $\pm$ 0.26 | 40.72 $\pm$ 0.31 | 21.79 $\pm$ 1.15 | 64.38 $\pm$ 3.51 | 35.59 $\pm$ 2.51 | 47.14 $\pm$ 0.29 |
| MAE, SSL baseline     | 1.00 $\pm$ 0.14 | 2.24 $\pm$ 0.29 | 5.30 $\pm$ 0.52  | 4.58 $\pm$ 0.17  | 31.34 $\pm$ 1.65 | 10.96 $\pm$ 0.44 | 11.75 $\pm$ 0.38 |
| MAE, with BRIDGE      | 1.38 $\pm$ 0.37 | 2.50 $\pm$ 0.70 | 7.54 $\pm$ 2.13  | 5.35 $\pm$ 0.76  | 35.58 $\pm$ 7.14 | 13.44 $\pm$ 3.46 | 15.46 $\pm$ 2.68 |